

AOS-CX 10.10 Quality of Service Guide

8360 Switch Series



**Hewlett Packard
Enterprise**

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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A, JL700C, JL701C, JL702C, JL703C, JL706C, JL707C, JL708C, JL709C, JL710C, JL711C, JL704C, JL705C, JL719C, JL718C, JL717C, JL720C, JL722C, JL721C)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <code><example-text></code> ■ <i>example-text</i> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Convention	Usage
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 83xx, 9300, and 10000 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.



If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

Quality of Service (QoS) enables network administrators to customize how different types of traffic are serviced on a network, taking into account the unique characteristics of each traffic type and its importance within an organization's infrastructure. QoS ensures uniform and efficient traffic handling, keeping the most important traffic moving at an acceptable speed, regardless of current bandwidth usage. It also provides methods for administrators to control the priority settings of inbound traffic arriving at each network device.

End-to-end QoS behavior

The QoS settings on each network device must be aligned to achieve the desired end-to-end QoS behavior for a network. Three service types can be used to categorize and prioritize network traffic:

- Best Effort Service
- Ethernet Class of Service (CoS)
- Internet Differentiated Services (DiffServ)

For a network as a whole, it is best to select one service type to use as the primary end-to-end behavior, and then use the other two service types as needed.

Best effort service

This is the simplest service type. All traffic is treated equally in a first-come, first-served manner. If the traffic load is low in relation to the capacity of the network links, then there is no need for the administrative complexity and costs of maintaining a more complex end-to-end policy. This is sometimes called over-provisioning, as all link speeds are much higher than peak loads on the network.

Class of Service

Class of Service (CoS) is a method for classifying network traffic at layer 2 by marking 802.1Q VLAN Ethernet frames with one of eight service classes.

CoS	Traffic type	Example protocols
7	Network Control	STP, PVST
6	Internetwork Control	BGP, OSPF, PIM
5	Voice (<10ms latency)	VoIP(UDP)
4	Video (<100ms latency)	RTP
3	Critical Applications	SQL RPC, SNMP

CoS	Traffic type	Example protocols
2	Excellent Effort	NFS, SMB
0	Best Effort	HTTP, TELNET
1	Background	SMTP, IMAP

CoS 1 is deliberately set as the lowest CoS. This enables a traffic service level below the default (best effort) traffic level to be specified.

The 3-bit Priority Code Point (PCP) field within the 16-bit Ethernet VLAN tag is used to mark the CoS.



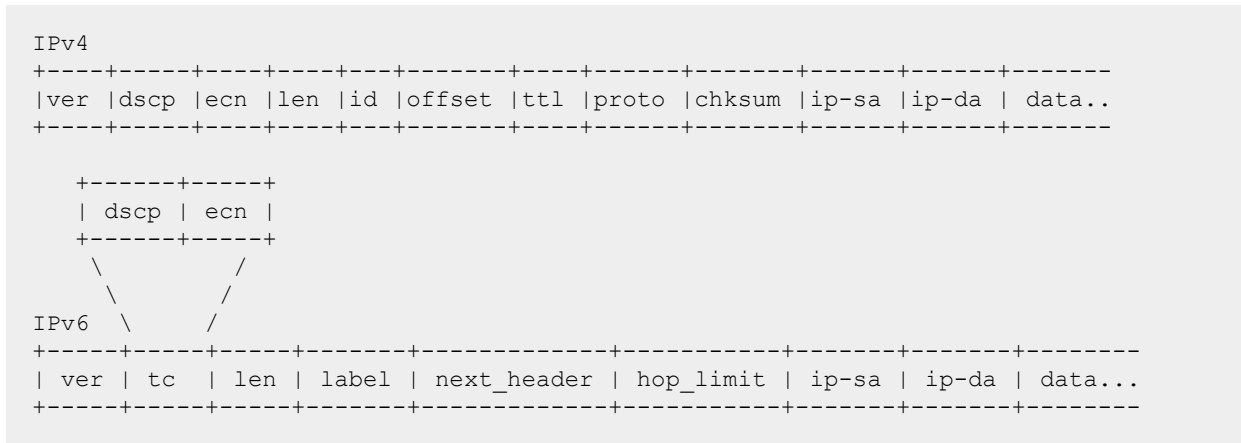
Differentiated services

Differentiated services (DiffServ) is a method for classifying network traffic at layer 3 by marking packets with one of 64 different service classes. Services classes are identified by the Differentiated services Code Point (DSCP) value. Some common DSCP values are:

DSCP	Name	Service class	RFC
48	CS6	Network Control	2474
46	EF	Telephony	3246
40	CS5	Signaling	2474
34, 36, 38	AF41, AF42, AF43	Multimedia Conferencing	2597
32	CS4	Real-Time Interactive	2474
26, 28, 30	AF31, AF32, AF33	Multimedia Streaming	2597
24	CS3	Broadcast Video	2474
18, 20, 22	AF21, AF22, AF23	Low-Latency Data	2597
16	CS2	OAM	2474
00	CS0,BE,DF	Best Effort	2474
10, 12, 14	AF11, AF12, AF13	Bulk Data	2597
08	CS1	Low-Priority Data	3662

DSCP CS1 (08) CoS 1 is deliberately set as the lowest priority. This enables a traffic service level below the standard (best effort or default forwarding) level to be specified.

The DSCP value is carried within the IPv4 DSCP field or the upper 6-bits of the 8-bit IPv6 Traffic Class (TC) field.



QoS on the switch

There are five key stages a packet passes through when traversing a switch: ingress, prioritization, destination determination, egress queuing, and transmission. The following table provides an overview of each stage, and lists the commands that can be used to configure QoS settings.

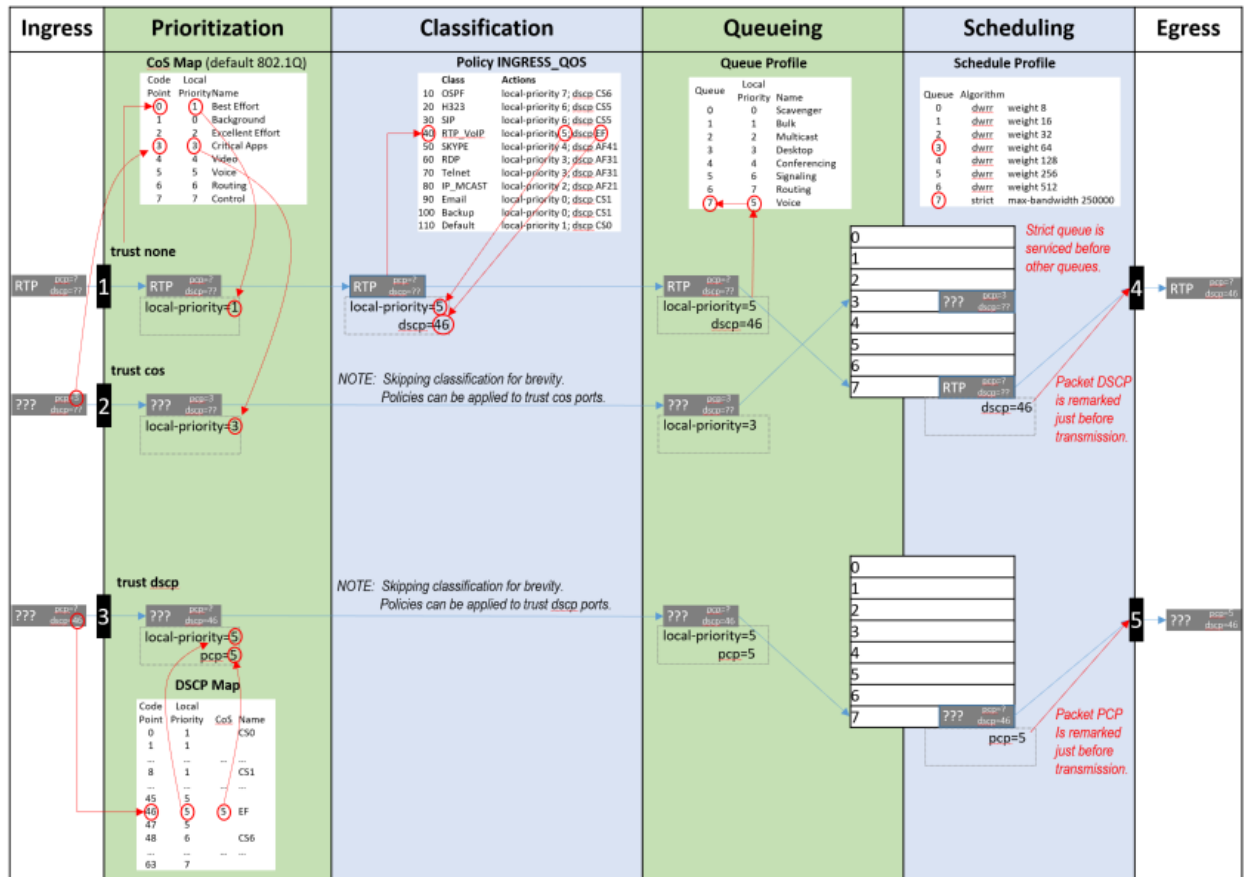


Switches with at least 52 ports will experience negative performance if a flood occurs where at least 42 ports are members of the same VLAN and all 52 ports have QoS rules applied to them.

Ingress	Prioritization	Classification	Queueing	Transmission
Packets arrive at switch interface (port).	An initial local-priority value is assigned to the packet based on VLAN CoS or IP DSCP.	Packets local-priority and code points can be remarked, and rates policed, based on many packet header fields. (see ACL & Policy Guide)	Packets are queued based on the destination interface and local-priority of the packet.	A scheduler defines the order in which packets are selected from queues to be transmitted from the interface.
rate-limit Rate limiting can control ingress flow by packet type: broadcast, multicast, unknown-unicast, or ICMP.	qos trust {cos dscp none} Assigns which packet values are used to determine initial local-priority either globally for all interfaces, or to specific interface(s). qos cos-map Defines how CoS values are mapped to local-priority values. qos dscp-map Defines how IP DSCP values are mapped to local-priority values. qos cos Remarks egress CoS value. qos dscp Remarks egress DSCP value.	class {ip ipv6 mac} NAME match TERMS [count] ignore TERMS [count] Creates an ordered list of rules to identify packets to match the class. policy NAME class NAME ACTIONS Creates an ordered list of classes to which traffic is evaluated, and the actions to be taken on matching packets: remarking, policing etc apply policy NAME Assigns the policy to evaluate all traffic inbound or outbound globally, or on specific interfaces or VLANs.	qos queue-profile Creates a profile that defines transmit queues. map queue Assigns a local-priority to a queue. Packets with a matching local-priority are placed in the queue. name queue Assigns a name to a queue. qos threshold-profile Creates a profile that defines ECN limits for queues. queue NUM action ecn Assigns an ECN limit for a queue. apply qos Assigns queue, schedule, or threshold profiles globally to all interfaces, or schedule or threshold profiles to a specific interface(s).	qos schedule-profile Creates a profile that defines the order in which packets are selected from the queues for transmission. strict queue Assigns the strict priority algorithm, with optional max-bandwidth, to a queue. dwrr queue Assigns the weighted DWRR or WFQ algorithm to a queue. qos shape Assigns the maximum rate for traffic to be transmitted by the interface.

Ingress	Prioritization	Classification	Queueing	Transmission
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	show int IFNAME qos Displays all the QoS settings configured on an interface: - Profiles - Rate-limits - CoS or DSCP remarks - Egress shape - Trust show qos cos-map Displays the contents of the CoS Map table. show qos dscp-map Displays the content of the DSCP Map table. show qos trust Display the current default trust mode.	show class {ip ipv6 mac} Displays the ordered list of the match and ignore statements in the class. show policy Displays the order list of classes in the policy. show policy hitcounts Displays the ordered list of classes in the policy. For each class the number of times that each match or ignore statement in the class matched a packet.	show int IFNAME queues Displays the per-queue counts of packets transmitted, packets dropped, bytes transmitted. show qos queue-profile Displays the definition of queue profiles. show qos threshold-profile Displays the definition of threshold profiles.	show qos schedule-profile Displays the definition of schedule profiles.

The following diagram shows how different packets might traverse a switch. It also shows how QoS configuration settings apply at each stage.



QoS trust

Traffic priorities for networks can be carried in VLAN tags, using the CoS Priority Code Point (PCP), or in IP packet headers, using the Differentiated Services Code Point (DSCP). Whether these priorities affect how traffic is serviced, depends on how QoS trust mode is configured on the switch. QoS trust mode specifies how the switch assigns local priority values to ingress packets. Trust mode can be set globally for all interfaces, or individually for each interface. By default, trust mode is set to `none`, meaning that any QoS information in the packet (CoS or DSCP) is ignored, and local priority values are assigned from the CoS map value for code point 0. An exception to this can be configured, allowing a QoS remark to be applied to DSCP values when trust mode is `none`.

When trust mode is set to CoS or DSCP, the switch translates the QoS settings in VLAN tags (for CoS), or the DS field in an IP header (for DSCP), to local priority values on the switch. Translation is controlled by the CoS map or DSCP map tables.

For example:

CoS Map

When QoS trust mode = CoS

Code Point	Local Priority	Name
0	1	Best Effort
1	0	Background
2	2	Excellent Effort
3	3	Critical Apps
4	4	Video
5	5	Voice
6	6	Routing
7	7	Control

Untagged packet → Packet local-priority = 1

Tagged PCP=3 packet → Packet local-priority = 3 & PCP = 3

DSCP Map

When QoS trust mode = DSCP

Code Point	Local Priority	CoS	Name
0	1		CS0
1	1		
...	...	...	...
8	1		CS1
...	...	...	...
45	5		
46	5	5	EF
47	5		
48	6		CS6
...	...	...	...
63	7		

Untagged IP DSCP=46 packet → Packet local-priority = 5 & PCP = 5 & DSCP = 46

Tagged PCP=3 IP DSCP=46 packet → Packet local-priority = 5 & PCP = 5 & DSCP = 46

When QoS trust mode = None

Tagged PCP=3 IP DSCP=32 packet	qos cos 6	Packet local-priority = 1 (Cos Map 0) & PCP = 6 & DSCP = 32
	qos dscp 48	Packet local-priority = 1 (Cos Map 0) & PCP = 3 & DSCP = 48

Dynamic QoS trust mode

The device profile feature can dynamically set the QoS trust mode on an interface based on the LLDP information exchanged with a link partner. The device profile's trust mode temporarily overrides the static trust mode configured for an interface. The override remains in place as long as that link partner is connected and its link state is **up**. Use command `show interface IFNAME qos` to view the current QoS trust mode for an interface.

Port rate limiting

Port rate limiting helps control undesirable traffic. Its purpose is to allow enough unicast, broadcast, multicast, and ICMP rate-limit traffic for the network to function properly, while preventing flooding and traffic storms.

A certain amount of each type of traffic is required for normal network operation. Broadcast packets may include ARP and DHCP traffic, for instance. Video streams, and certain types of network protocol packets, are multicasts. Unknown-unicast packets may be intended for devices whose addresses have temporarily aged out of network-forwarding caches. Configuring rate limits can help provide the balance between necessary and flooded traffic.

Queue profiles

A queue profile defines the queues that are associated with an interface to control the transmission of packets. Each profile supports up to eight queues, numbered 0 to 7. The larger the queue number, the

higher its priority during transmission scheduling. Packets are assigned to a queue based on their local priority value (0 to 7). A queue profile must map all eight local priority values to whatever queues are being used on the switch, and a schedule profile must specify the configuration for those same queues. A queue without a local priority value assigned to it is not used to store packets.

The switch is automatically provisioned with an initial queue profile named `factory-default` which assigns each local priority to the queue of the same number. To see the default queue profile, use the command `show qos queue-profile factory-default`:

```
switch# show qos queue-profile factory-default
queue_num local_priorities name
-----
0          0                  Scavenger_and_backup_data
1          1
2          2
3          3
4          4
5          5
6          6
7          7
```

More than one local priority value can be assigned to the same queue. For example,

Local Priority	Queue
0	0
1	1
2	2
3	3
4	4
5	5
6	5
7	5

Commonly used commands for working with QoS queues are as follows:

- `qos queue-profile`: Creates an empty queue-profile and enters the profile configuration context.
- `name queue`: Assigns a descriptive name to a queue.
- `map queue`: Assigns a local-priority to a queue.
- `apply qos queue-profile`: Applies a queue-profile globally to all interfaces.

Schedule profiles

A schedule profile determines the order in which queues are selected for transmission, and the amount of service available for each queue. A schedule profile must be configured on every interface at all times. A schedule profile can be applied globally to all interfaces, or only to specific interfaces.

Three options are available:

- All queues use deficit weighted round robin queuing (DWRR)
- All queues use strict priority
- The highest priority queue uses strict priority, and all other queues use DWRR



NOTE: On the 8360 series switch all queues use guaranteed minimum bandwidth

A weighted schedule profile assigns relative servicing for each queue. The amount of service per weight is relative to the underlying hardware implementation, and to the weights assigned to the other non-empty queues. Strict scheduling can be used to service queues purely on the basis of highest priority first (at the risk of starving lower-priority queues during high stress periods). A combination of strict and weighted scheduling offers more service to the highest priority queue when needed, while preserving scheduling between the remaining queues, thus decreasing the risk of starvation.

The switch is automatically provisioned with a schedule profile named **factory-default**, which assigns DWRR to all queues with a weight of 1. Use the command `show schedule-profile factory-default` to view the default schedule profile. (Do not use `show running-configuration`, as it only displays changes from the initial settings.)

```
switch# show qos schedule-profile default
queue_num algorithm weight
-----
0          dwrr       1
1          dwrr       1
2          dwrr       1
3          dwrr       1
4          dwrr       1
5          dwrr       1
6          dwrr       1
7          dwrr       1
```

Egress queue shaping

Egress queue shaping limits the amount of traffic transmitted per strict output queue. The buffer associated with each egress queue stores excess traffic to absorb bursts and smooths the output rate. For example, an administrator might limit strict-priority queue traffic to prevent low-priority queue starvation in the event that a device inappropriately sends too many higher-priority packets.

Egress queue shaping can be configured on an Ethernet port or on a link aggregation group (LAG). To configure egress queue shaping, define a schedule profile with the strict priority algorithm assigned to each queue.

Egress port shaping

Egress port shaping limits the amount of aggregate traffic transmitted through a port. To be effective, the egress port-shaping rate must be less than the port's line rate. By default, the egress port-shaping rate is the same as the line-rate of the port. Buffers associated with each port store excess traffic. When both egress port-shaping and egress queue-shaping are configured on the same interface, the switch respects the minimum of both configurations.

Explicit Congestion Notification

Explicit Congestion Notification (ECN) provides a mechanism for two end-points to exchange end-to-end notification of network congestion. ECN uses a 2-bit field in the IP header to indicate that the traffic load

on network equipment in the path between an ECN-capable sender and receiver is causing packets to be buffered, as defined by IETF RFC 3168 (<https://tools.ietf.org/html/rfc3168>).

Threshold profiles

Threshold profiles configure individual queue utilization thresholds as triggers for taking action (i.e., ECN marking) on a packet. A threshold profile is applied per-port and defines the threshold and action for each queue. Omitting configuration for a queue in a threshold profile means that queue will not be configured with a threshold value or action.

In an environment where responsive transport protocols are in use and congestion management features are required to reduce latency, ECN can be configured on queues carrying delay-sensitive traffic. The result is that queue utilization is actively managed, resulting in ECT packets being CE marked when queue utilization reaches or exceeds a configured threshold.

Terms

Class

For networking, a set of packets sharing a common characteristic. For example, all IPv4 packets.

Code point

The name of a packet header field, or the value carried within a packet header field:

- Example 1: Priority code point (PCP) is the name of a field in the IEEE 802.1Q VLAN tag.
- Example 2: Differentiated services code point (DSCP) is the name of a field carried within the DS field of an IP packet header.

Color

A metadata label associated with each packet within the switch. It has three values: green (0), yellow (1), or red (2). When packets encounter congestion for a resource (queue), the switch uses packet color to distinguish which packets must be dropped, and is mostly used for packets marked with Assured Forwarding (AF) DSCP values.

Not supported in this release.

Class of service (CoS)

A 3-bit value used to mark packets with one of eight classes (levels of priority). It is carried within the priority code point (PCP) field of the IEEE 802.1Q VLAN tag.

Differentiated services code point (DSCP)

A 6-bit value used to mark packets for different per-hop behavior as originally defined by IETF RFC 2474. It is carried within the differentiated services (DS) field of the IPv4 or IPv6 header.

Local priority

A meta-data label associated with a packet within the switch which is used to classify packets for different treatment (such as queue assignment). Eight local priorities are defined on the switch, numbered from 0 to 7. A queue profile must map all eight local priorities to whatever queues are in use on the switch, and a schedule profile must specify the configuration for these same queues.

Metadata

Information labels associated with each packet in the switch, separate from the packet headers and data. These labels are used by the switch in its handling of the packet. For example: arrival port, egress port, VLAN membership, and local priority.

Priority code point (PCP)

The name of a 3-bit field in the IEEE 802.1Q VLAN tag. It carries the CoS value to mark a packet with one of eight classes (priority levels).

Quality of service (QoS)

General term used when describing or measuring performance. For networking, it means how different classes of packets are treated when traversing a network or device.

Traffic class (TC)

General term for a set of packets sharing a common characteristic. It used to be the name of an 8-bit field in the IPv6 header originally defined by IETF RFC 2460. This field name was changed to differentiated services by IETF RFC 2474.

Type of service (ToS)

General term when there are different levels of treatment (fare class). It used to be the name of an 8-bit field in the IPv4 header originally defined by IETF RFC 791. This field name was changed to differentiated services by IETF RFC 2474.

Configuring QoS

Procedure

1. Optionally, add a rate limit for ingress traffic on one or more interfaces with the command `rate-limit`.
2. If you do not want to use the default QoS queue profile to map local priority to queue, create one or more custom queue profiles with the command `qos queue-profile`. For each queue in a custom queue profile:
 - a. Assign a local priority value with the command `map queue`.
 - b. Optionally, define a descriptive name with the command `name queue`. All local priorities (0 to 7) must be mapped to a queue, and the queues selected for use must be in contiguous order starting at 0.
3. If you do not want to use the default QoS schedule profile to determine the order in which queues are selected to transmit a packet, create one or more custom schedule profiles with the command `qos schedule-profile`. For each queue in a custom schedule queue profile, define scheduling priority with the commands `strict queue` and `dwrr queue`.
4. Optionally for strict queues, configure egress queue shaping to limit egress bandwidth on an interface to a value that is less than its line rate. Use the `max-bandwidth` parameter of the `strict queue` command.
5. Activate QoS settings with the command `apply qos`. This command lets you apply a queue profile and schedule profile globally to all interfaces, or a schedule profile override to individual interfaces.
6. View QoS configuration settings with the provided `show` commands.

Configuring expedited forwarding for VoIP traffic

Voice over IP (VoIP) traffic is delay and jitter sensitive. For optimum transmission of VoIP traffic, dwell time in network devices must be kept to a minimum and all network devices in the data path must have identical per-hop behaviors. To configure a dedicated queue on the switch to handle VoIP traffic with priority service before all other queues, follow these steps.

Prerequisites

This scenario assumes that VoIP packets are uniquely identified using DiffServ code point 46, Expedited Forwarding (EF).

Procedure

1. Map DSCP EF packets exclusively to local priority 5. The default DSCP map has eight code points (40 through 47), that are mapped to local priority 5. To reserve local priority 5 for VoIP traffic, the

other code points must be reassigned. In this scenario, local priority 6 is used for all reassignments, including for code point 40, Call Signaling protocol (CS5).

```
switch(config)# qos dscp-map 40 local-priority 6 name CS5
switch(config)# qos dscp-map 41 local-priority 6
switch(config)# qos dscp-map 42 local-priority 6
switch(config)# qos dscp-map 43 local-priority 6
switch(config)# qos dscp-map 44 local-priority 6
switch(config)# qos dscp-map 45 local-priority 6
switch(config)# qos dscp-map 47 local-priority 6
```

2. Queue 7 is the highest priority queue, so for best throughput, create a queue profile that maps local priority to queue 7.

```
switch(config)# qos queue-profile ef_priority
switch(config-queue)# name queue 7 Voice_Priority_Queue
switch(config-queue)# map queue 7 local-priority 5
switch(config-queue)# map queue 6 local-priority 7
switch(config-queue)# map queue 5 local-priority 6
switch(config-queue)# map queue 4 local-priority 4
switch(config-queue)# map queue 3 local-priority 3
switch(config-queue)# map queue 2 local-priority 2
switch(config-queue)# map queue 1 local-priority 1
switch(config-queue)# map queue 0 local-priority 0
```

3. Apply the profiles to all interfaces.

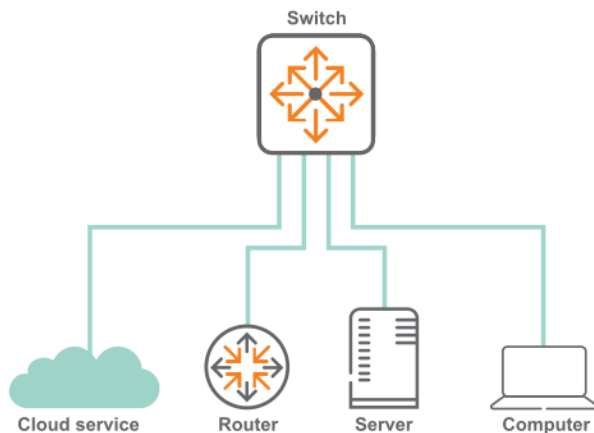
```
switch(config)# apply qos queue-profile ef_priority schedule-profile voip
```

4. Configure DSCP trust mode on all ports

```
switch(config)# qos trust dscp
```

Configuring rate limiting

This scenario illustrates how to use rate limiting to manage the traffic from various devices connected to a switch. The physical topology of the network looks like this:



A certain amount of broadcast traffic is necessary to maintain healthy network operation, particularly from routers and across service boundaries. In this scenario, both the service cloud and the router connections limit this traffic to 1 Gbps. The server has a smaller limit, as it does not require as much network protocol traffic as the service cloud and router.

A multicast server needs to be able to stream multicast traffic to clients, so a multicast rate limit may not be helpful. A computer, however, should not be generating large amounts of multicast traffic (it may be receiving streams, but typically not sending them). In this example, the computer is configured with a multicast rate limit to prevent malicious traffic from taking up network bandwidth.

Finally, while the service cloud and router may need to send traffic for unknown unicast addresses to resolve address forwarding, the server and computer should send very little of this type of traffic. Rate limiting unknown unicast traffic on those two devices enforces that.

Procedure

1. Configure broadcast and multicast rate limiting for the service cloud connection.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# rate-limit broadcast 500 pps
switch(config-if)# rate-limit multicast 500 pps
switch(config-if)# exit
```

2. Configure broadcast rate limiting for the router connection.

```
switch(config-if)# interface 1/1/2
switch(config-if)# rate-limit broadcast 500 pps
switch(config-if)# exit
```

3. Configure broadcast rate limiting for the server connection.

```
switch(config-if)# interface 1/1/5
switch(config-if)# rate-limit broadcast 100 pps
switch(config-if)# exit
```

4. Configure broadcast, and multicast rate limiting for the computer connection.

```
switch(config-if)# interface 1/1/10
switch(config-if)# rate-limit broadcast 50 pps
switch(config-if)# rate-limit multicast 50 pps
```

Configuring egress queue shaping

This example shows how to apply egress queue shaping to an interface. First, a schedule profile is created that has per-queue bandwidth limits set on all queues with `strict` as the scheduling algorithm. Next, this profile is applied to an interface or LAG.

The following example creates a schedule profile named **EQSExample**, which services all seven queues using `strict` priority. This profile configures queues 1, 4, and 7 with a bandwidth limit of 10 Gbps, 20 Gbps, and 30 Gbps respectively. The profile is then applied to interface 1/1/1.

```

switch(config)# qos schedule-profile EQSEExample
switch(config-schedule)# strict queue 0
switch(config-schedule)# strict queue 1 max-bandwidth 10000000
switch(config-schedule)# strict queue 2
switch(config-schedule)# strict queue 3
switch(config-schedule)# strict queue 4 max-bandwidth 20000000
switch(config-schedule)# strict queue 5
switch(config-schedule)# strict queue 6
switch(config-schedule)# strict queue 7 max-bandwidth 30000000
switch(config-schedule)# exit
switch(config)# interface 1/1/1
switch(config-if)# apply qos schedule-profile EQSEExample

```

Configuring egress port shaping

This example shows how to apply egress port shaping to an interface to limit the rate of egress traffic. Egress port shaping is configured by specifying the desired bandwidth rate in kilobits per second (kbps). To be effective, the egress rate must be less than the line rate of the egress interface. If the configured egress rate exceeds the interface's line rate, then egress shaping has no effect.

The configured egress rate on a specific interface can be found by using the `show interface` and `show interface qos` commands.

The following example configures an egress rate of 100 Mbps .

```

switch(config)# interface 1/1/1
switch(config-if)# qos shape 100000

```

In the next example, both egress port shaping and egress queue shaping are configured on the same interface.

The example creates a schedule profile named `EQSEExample` with strict priority for all seven queues. Queue 7 is configured with a bandwidth limit of 300 Mbps. The profile is then applied to interface `1/1/1` with egress port shaping of 400 Mbps . As egress queue shaping and egress port shaping are both configured on port `1/1/1`, egress queue shaping is subject to the lower port or queue shape rate. The effective bandwidth for the traffic egressing on queue 7 will be 300 Mbps .

```

switch(config)# qos schedule-profile EQSEExample
switch(config-schedule)# strict queue 0
switch(config-schedule)# strict queue 1
switch(config-schedule)# strict queue 2
switch(config-schedule)# strict queue 3
switch(config-schedule)# strict queue 4
switch(config-schedule)# strict queue 5
switch(config-schedule)# strict queue 6
switch(config-schedule)# strict queue 7 max-bandwidth 300000
switch(config-schedule)# exit
switch(config)# interface 1/1/1
switch(config-if)# apply qos schedule-profile EQSEExample
switch(config-if)# qos shape 400000

```

Supporting Ethernet 802.1D Class of Service

IEEE 802.1Q is the most current Ethernet standard for Class of Service (CoS). It superseded an earlier standard, 802.1D, in 2005. IEEE 802.1Q slightly changed the ordering of the classes of service from its

predecessor IEEE 802.1D for CoS 2 and CoS 0:

CoS 802.1Q	CoS 802.1D
7 Network Control	7 Network Control
6 Internetwork Control	6 Voice (<10ms latency)
5 Voice (<10ms latency)	5 Video (<100ms latency)
4 Video (<100ms latency)	4 Controlled Load
3 Critical Applications	3 Excellent Effort
2 Excellent Effort	0 Best Effort
0 Best Effort	2 Spare
1 Background	1 Background

In 802.1D, both CoS 2 and CoS 1 are below CoS 0 (Best Effort).

When a switch is installed in a network of devices following 802.1D Class of Service, the QoS Cos map must be re-configured to follow the 802.1D standard by swapping the assignments of CoS 0 and 2:

```
switch# config
switch(config)# qos cos-map 0 local-priority 2 color green name Best_Effort
switch(config)# qos cos-map 2 local-priority 1 color green name Spare
switch(config)# exit
switch# show qos cos-map
code_point  local_priority  color    name
-----
0           2             green    Best_Effort
1           0             green    Background
2           1             green    Spare
3           3             green    Critical_Applications
4           4             green    Video
5           5             green    Voice
6           6             green    Internetwork_Control
7           7             green    Network_Control
```

Monitoring queue operation

Use the show interface queues command to display the traffic transmitted per queue, and the number of packets dropped due to the queue being full. For example:

```
switch# show interface 1/1/1 queues
Interface 1/1/1 is (Administratively down)
Admin state is down
State information: admin_down
```

	Tx Packets	Tx Bytes	Tx Drops
Q0	100	8000	0
Q1	1234567	12345678908	5
Q2	0	0	0
Q3	0	0	0
Q4	0	0	0
Q5	0	0	0
Q6	0	0	0
Q7	0	0	0

- **Tx Bytes:** Total bytes transmitted. The byte count may include packet headers and internal metadata that are removed before the packet is transmitted. Packet headers added when the packet is transmitted may not be included.
- **Tx Packets:** Total packets transmitted.
- **Tx Drops:** The number of packets dropped by a queue before it was sent. When traffic cannot be transmitted out of an egress interface, it may be buffered in memory if space is available. The more servicing assigned to a queue by a schedule profile, the less likely traffic destined for that queue will back up and potentially dropped.

Data center bridging for storage and lossless Ethernet



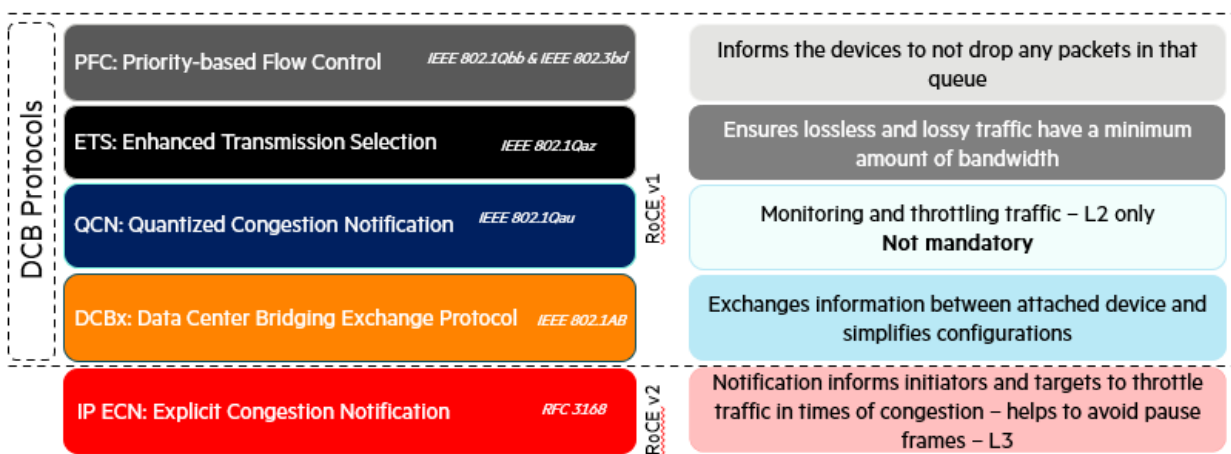
Supported on the 8360 Switch Series.

The term *lossless Ethernet* is intertwined with the Data Center Bridging protocol suite (DCB) to eliminate traffic loss within an Ethernet fabric. This traffic loss is achieved via buffer queue overflow for selected traffic profiles. DCB protocols enable different priorities for specific traffic flows to avoid packet drops, head of line blocking, and excessive latency across an Ethernet fabric.

DCB achieves this by managing bandwidth, priority, and flow control of selected traffic types when sharing the same Ethernet network link.

Data center bridging components

Data center bridging is comprised of the following key components:



IP ECN is an extension to IP supporting TCP and is covered by RFC 3168. It is a requirement to support lossless traffic flows over L3 networks.

PFC - Priority-based flow control

Priority-based Flow Control (PFC) is a link-level flow control mechanism similar to Ethernet PAUSE and standardized in IEEE 802.1Qbb.

The standard Ethernet PAUSE stops all traffic on an Ethernet link impacting all traffic flows. PFC enables the division of traffic on a link into eight priorities and pauses the traffic of a selected priority under congestion without stopping the traffic assigned to other priorities on the link.

ETS - Enhanced transmission selection

Enhanced Transmission Selection (ETS) allows management of the link bandwidth by specifying bandwidth availability to each priority group. Priorities can be grouped according to bandwidth or bandwidth allocated to each individual priority.

QCN - Quantized congestion notification

Quantized Congestion Notification (QCN) provides congestion control at Layer 2. It was developed in parallel to the related effort on PFC.

QCN is designed to provide congestion control for long-lived flows in limited bandwidth delay environments and is applied solely to a single layer 2 domain.



This standard is not required if the PFC feature is available and is not supported on the CX switch platform.

DCBx - Data center bridging exchange protocol

Data Center Bridging Exchange protocol (DCBx) is a discovery and capability exchange protocol to discover peers and negotiate Data Center Bridging configuration. DCBx is specified as part of IEEE 802.1Qaz-2011. DCBx uses LLDP as the underlying protocol for exchange of parameters with the peer. The DCBx parameters are exchanged as LLDP TLVs. AOS-CX switches support both versions of DCBx: IEEE DCBx and CEE DCBx.

DCBx supports VSX synchronization. For more information about enabling VSX synchronization, see the *Virtual Switching Extension (VSX) Guide* for your switch and software version.

DCBx is an extension of LLDP used by DCB devices to exchange configuration information with directly connected peers. Its applications include:

- Discovery of peer capabilities
- Detection of misconfigurations and identifying configuration mismatches
- Advising a DCB peer of which DCB configurations it should adopt

DCBx guidelines

- DCBx is disabled by default.
- LLDP must be enabled on the interfaces supporting DCBx.
- DCBx should be configured on all L2 interfaces.
- L3 hops which leverage PFC and ECN do not require DCBx.
- DCBx is only supported on physical interfaces and not on management or logical interfaces, similar to how LLDP behaves.
- DCBx is not essential between host and switch but is highly desirable as to avoid manual PFC configurations of host devices.
- Supported protocols include:
 - DCBx Converged Enhanced Ethernet (CEE)
 - DCBx IEEE 802.1Qaz
 - AOS-CX supports only the IEEE 802.1Qaz 2011 version of DCBx with an OUI of 0x0080c2.
- AOS-CX advertises DCBx with the willing bit set to 0 in all TLVs. This tells the peer that the switch is not willing to change its configuration to match the peer's configuration.
- When a peer switch does not support IEEE DCBx, a misconfiguration error will be displayed in the `show dcbx interface output`.

IP ECN

IP Explicit Congestion Notification (ECN) is not part of the DCB protocol standard and is covered separately by RFC 3168, however ECN plays a critical role in enhancing lossless Ethernet over a layer 3 IP fabric. IP ECN is a congestion notification protocol which throttles back traffic rather than pausing or dropping packets in the event of congestion. It is a slow acting mechanism and assists in avoiding pause frames generated by PFC by throttling back traffic before queue congestion occurs.

When devices in an IP ECN fabric experience congestion, the device marks congestion via ECN bits in the IP DiffServ field and forwards the IP packets on to either the next IP hop or to the end host destination. The receiving host is IP ECN aware and the congestion notification is handled by the upper layer protocols such as TCP. The IP ECN congestion bits are then echoed back to the transmitting host device which in turn reduces its transmission rate.

In order for IP ECN to work, the sender and receiver's upper layer protocols must use a protocol that is congestion-aware such as TCP or SCTP (Stream Control Transmission Protocol). In addition, intermediate switch hops are required to be ECN-aware and must be configured appropriately.

Host network interface cards

Network interface cards (NICs) on hosts that require lossless Ethernet can support the following protocols:

- PFC
- ETS
- DCBx*
- IP ECN



*Support for DCBx is not mandatory but highly desirable.

PFC support is a prerequisite for storage and lossless Ethernet networking. AOS-CX provides the following DCB features which are supported across specific listed CX platforms:

Feature		AOS-CX 10.10
DCB Protocols	Priority-based Flow Control (PFC)	Yes (10000/9300/8400/8360/8325)
	Enhanced Transmission Selection (ETS)	Yes (10000/9300/8400/8360/8325)
	Data Center Bridging Exchange (DCBx)	Yes (10000/9300/8400/8360/8325)
	Quantized Congestion Notification (QCN)	No
L3 Notification Protocol	IP Explicit Congestion Notification (ECN)	Yes (10000/9300/8400/8360/8325)
Lossless PFC priorities per port		7 (10000/9300/8325) 2 (8360) 1 (8400)
Lossless pools per port		3 (10000/9300/8325)

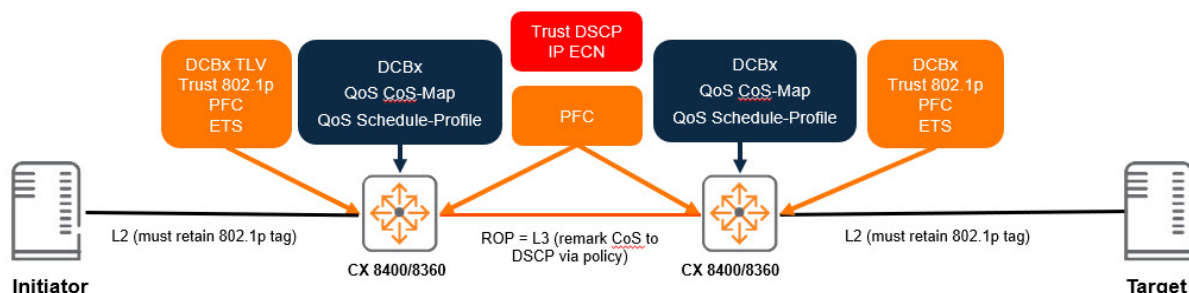


DCB protocols are only supported on the network underlay and are not supported across VXLAN overlay.

DCB layer 3 configuration task list

1. Enable DCBx
2. Configure and apply QoS queue and schedule profiles
3. Configure global trust and override trust settings on individual interface settings
4. Configure PFC
5. Configure application TLV with LLDP DCBx
6. Configure ECN

↗



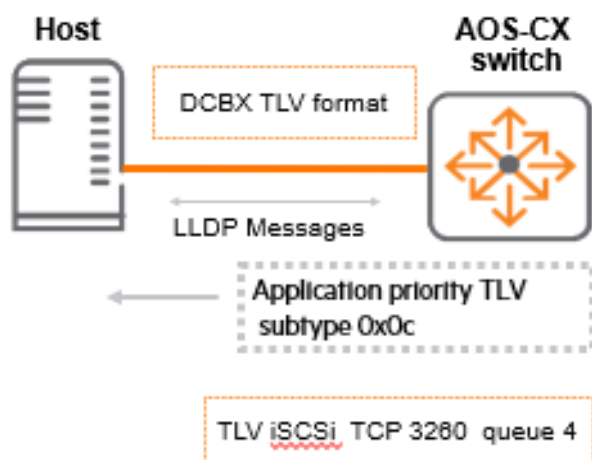
DCBx configuration

DCBx is dependent on LLDP and uses LLDP TLVs to exchange configuration details. DCBx uses a specific LLDP PDU TLV subtype of 0x0c. A DCBx-enabled device can advertise various parameters including ETS, PFC, 802.1p CoS value.

A DCBx-enabled switch can request its host peer to map certain application traffic to a priority using the DCBx TLV subtype (0x0c). Whether the host accepts the QoS parameters or rejects them is dependent on the host's local DCBx NIC willing "W" bit state:

- If set to 1 - Indicates that the adapter is willing to accept QoS parameters from the remote peer
- If set to 0 - Operational QoS parameters are always resolved from the local Host nic QoS parameters

DCBx TLV advertisement where the switch is advertises to the host to use the TCP iSCSI port 3260 and places it in queue 4 (802.1p CoC value 4):



DCBx configuration considerations

DCBx is expected to operate over a point-to-point link. If multiple LLDP neighbors are detected, then DCBx behaves if its peer DCBx TLVs are not present and behaves this way until the multiple LLDP neighbor condition is no longer present.

- DCBx can be enabled globally or within a specific interface and is disabled by default
- The IEEE DCBx IEEE 802.1Qaz is the default version when DCBx is enabled
- LLDP must be enabled prior to configuring DCBx
- DCBx is not essential between host and switch but is highly desirable as to avoid manual PFC configurations of host devices.
- DCBx should be configured on every switch hop interface.

Enabling DCBx

LLDP must be enabled globally prior to any DCBx configuration and can be enabled with the `lldp enable` command. DCBx can be enabled either globally or at the interface level as well as with the IEEE and CEE versions by using the `lldp dcbx` command.



The DCBx application TLV is part of the DCBx configuration but must be configured after the PFC configuration.

Enabling DCBx globally:

```
switch(config)# lldp dcbx
```

Enabling the IEEE version of DCBx globally:

```
switch(config)# lldp dcbx version ieee
```

Enabling the CEE version of DCBx globally:

```
switch(config)# lldp dcbx version cee
```

Enabling DCBx for an interface:

```
switch(config-if)# lldp dcbx
```

Enabling the IEEE version of DCBx for an interface:

```
switch(config-if)# lldp dcbx version ieee
```

Enabling the CEE version of DCBx for an interface:

```
switch(config-if)# lldp dcbx version cee
```

The `no` form of any of these commands will remove from the switch configuration. In order to do this you must first enter the appropriate configuration context as indicated by the `switch(config)#`

prompt for global configuration or `switch(config-if)#` for interface configuration. For example, removing the IEEE version of DCBx from a specific interface:

```
switch(config-if)# no lldp dcbx version ieee
```

Priority-based flow control

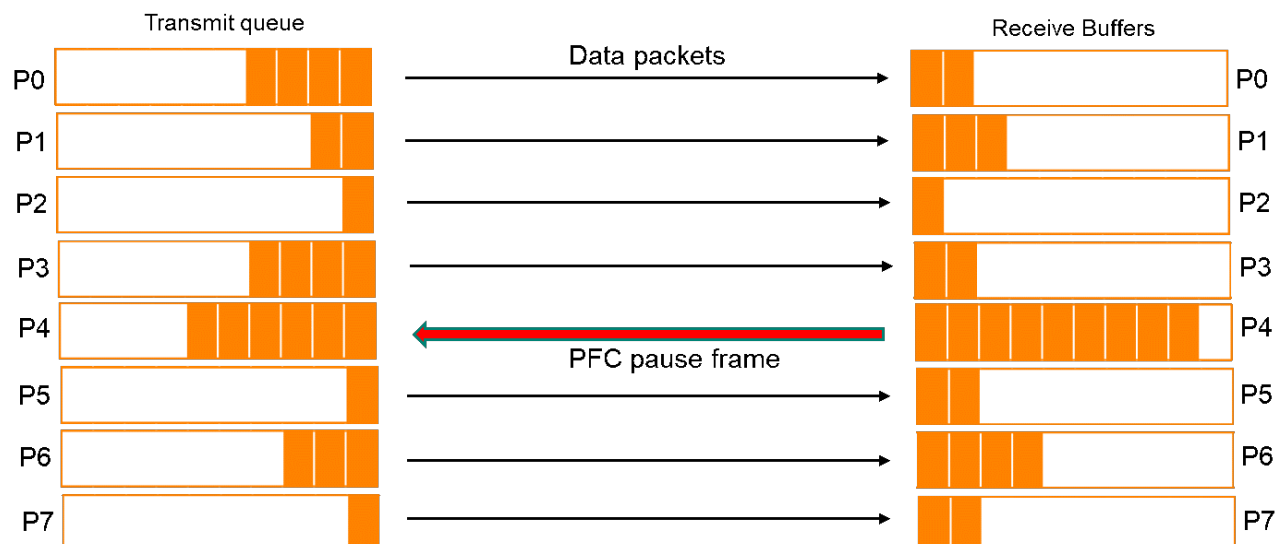
Priority-based Flow Control (PFC) provides an enhancement to the Ethernet flow control pause command. PFC operates at layer 2, supporting layer 2 and layer 3 network connectivity. The standard Ethernet pause frame will pause all traffic on an Ethernet link when an ingress interface buffer is full and impacting all traffic flows.

PFC provides 8 different traffic classes which can be aligned to up to eight queues and multiple traffic classes can also be assigned to a single queue. The PFC receiving side sends a pause frame (XOFF) to the transmitting station before the buffer becomes full for a specific traffic class without impacting other traffic classes.

Switch series	Priority-based flow control priorities
8325/9300/10000	Up to 7 PFC priorities per interface.
8360	Up to 2 PFC priorities per interface.
8400	One PFC priority configured per interface.

PFC has the following characteristics:

- IEEE 802.1Qbb is based on Ethernet flow control
- Pause frames stop transmitters when upstream buffer nearly full
- Required to support lossless Ethernet (avoid dropping storage traffic)
- Provides mechanism with PAUSE per 802.1p priority
- Can provide PAUSE of specific traffic classes only
- Jumbo frame MTU size is required on network infrastructure to support storage networking traffic



IP explicit congestion notification

TCP/IP networks signal congestion by dropping packets. This type of congestion is easily compensated using TCP/IP and is not tolerated for lossless network flows across Ethernet fabric. Instead of dropping packets when congestion is experienced, ECN marks the IP packet as congestion experienced (CE).

ECN is used in conjunction with DCBx and used to avoid hitting the PFC limits over L3 network hops. Packets are marked with ECN bits by each switch at a configurable ECN threshold which allows TCP to function normally without dropping IP packets. ECN has the following characteristics:

- Uses the DiffServ field in the IP header to mark the congestion status along the packet transmission path.
- End-to-end communication protocol which requires the transmitting and receiving device along with the intermediate switch hops to support it.
- Applied on a queue basis and can be applied to 1 or more queues concurrently up to the 8 queue limit.

ECN uses the two "least significant" bits in the Traffic Class field of the IP header (the bits that are farthest to the right) to signal the current state of network congestion. The possible markings are:

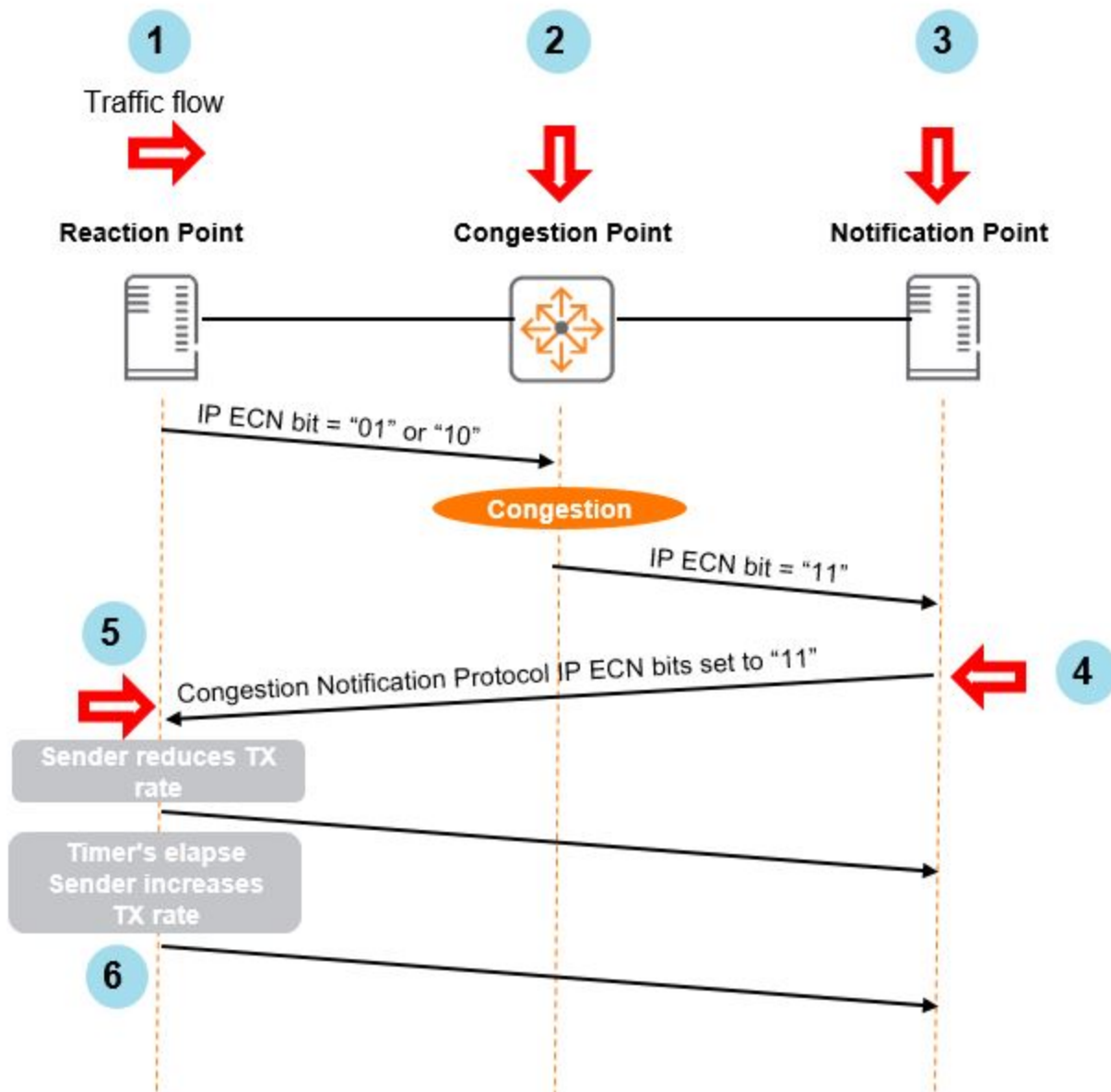
Bits	Congestion status	Abbreviation
00	Non ECN-Capable Transport	Non-ECT
01	ECN Capable Transport	ECT(1)
10	ECN Capable Transport	ECT(0)
11	Congestion Encountered	CE



ECT(1) and ECT(0) are considered to be functionally the same by the switch. If congestion is encountered, the zero bit will be flipped to indicate congestion.

ECN prevents dropped packets via the following actions:

1. IP packets flow normally on an IP network from an ECN-enabled device with the ECN field marked as uncongested (01 or 10)
2. If an ECN-enabled host device receives a packet marked 11 or CE, it acknowledges the marked packet with its upper layer protocols (TCP)
3. The host receiving the CE marked packet sends the congestion notification back to the transmitting host
4. The transmitting host device acknowledges receipt of the ECN congestion markings and reduces its transmission rate
5. The transmitting device increases the transmission rate after a timer elapses



IP ECN configuration considerations

The Aruba 8360 Switch Series has a slope threshold which is calculated as a percentage. When this threshold is met, the packets are immediately marked for ECN congestion. There is no upper threshold limit. Because the packets are marked for congestion when the threshold is met, setting the congestion threshold too low will impair network performance as the end host's TCP flow backs off when it receives the ECN congestion marked bits in the IP packet. Conversely, setting the IP ECN threshold too high may trigger PFC pause between network switches and end hosts.

The following factors should always be taken into account when configuring IP ECN:

- ECN is configured on queues carrying delay-sensitive traffic.
- Any number of queues maybe configured in a profile.
- An unconfigured queue means no threshold action behavior will occur.
- ECN can be applied globally or on individual interfaces.
- No threshold policies are applied by default.

- All switches in the transmission path between two ECN-enabled endpoints must have ECN enabled and configured for the appropriate queue.
- Testing is recommended to ensure ECN thresholds are correctly applied.

IP ECN configuration procedure

The procedure for configuring IP ECN is as follows:

- Create a threshold profile using the [qos threshold-profile](#) command and apply threshold values
- Apply the qos threshold profile globally
- Apply the qos threshold profile at the desired interface to override the global threshold setting

Creating a QoS threshold profile called **ECN** applied globally to all interfaces:

```
switch(config)# qos threshold-profile ECN
switch(config-threshold)# queue 1 action ecn all threshold 50 percent
```

Creating a QoS threshold profile called **ECN2** :

```
switch(config)# qos threshold-profile ECN2
switch(config-threshold)# queue 1 action ecn all threshold 60 percent
```

Applying QoS threshold profile **ECN2** to a specific interface:

```
switch(config)# interface 1/1/1
switch(config-if)# apply qos threshold-profile ECN2
```

Lossless QoS pool

The QoS pool is a packet buffer pool. The Aruba 8325 and 10000 Switch Series have a configurable QoS pool with an associated headroom buffer space, while the 8360 and 8400 have a single fixed QoS pool which is not exposed to the CLI for configuration.

Switch series	Lossless QoS pools
8325/9300/10000	Three configurable lossless pools
8360/8400	Single fixed lossless pool

The QoS pool is a packet buffer pool resident in memory in the switch. Only the QoS pools on the Aruba 8325 and 10000 Switch Series are configurable and have the following attributes:

- The QoS pool option enables the creation of a packet buffer pool which is dedicated to lossless traffic
- The QoS pool command allows the creation of a lossless pool size, headroom buffer associated with the pool, and the priorities that are mapped to that pool.
- The lossless pool size is a percentage of the total available buffer memory on the device.
- The headroom pool memory is allocated from the lossless pool and is used for storing packets that arrive on a port after a pause has been asserted.

Considerations and prerequisites

There are no prerequisite configuration activities required to configure QoS pools. However, a PFC priority or any related PFC commands cannot be enabled on an interface until the creation of the QoS pool and the appropriate PFC priority mapping to that pool. The following considerations should be taken into account when configuring QoS pools:

- Maximum buffer size in memory: 32768 kbytes (32Mb)
- Maximum configurable headroom memory size: 10240 kbytes (10Mb)
- Factory-default headroom size: 3072 kbytes (3Mb)
- The sum of pool sizes cannot be larger than 90% of the available Kbyte memory
- The headroom size cannot be greater than half of the lossless pool buffer size.



Best practice for QoS pools on all platforms is to not align the best effort traffic class (default local priority 1) and the critical control traffic (local priority 7) to a lossless pool.

QoS pool configuration

QoS pools are enabled using the QoS pool command in the global configuration context.



For additional information on QoS pool configuration please refer to the `qos pool` command.



The following examples are applicable to the 8325, 9300, and 10000 Switch Series.

Configuration options for QoS pool size:

```
switch(config)# qos pool <1-3> lossless size ?  
  <10-90>           The percentage of packet buffer memory to be allocated to  
                    this pool in integer format  
  <10.00-90.00>     The percentage of packet buffer memory to be allocated to  
                    this pool in decimal format  
  factory-default   Configure the pool size to the default value
```

Configuration options for headroom size:

```
switch(config)# qos pool 2 lossless size factory-default percent headroom ?  
  <0-10240>         The headroom buffer size in kilobytes  
  factory-default   Configure the default headroom buffer size in kilobytes
```

Configuring a QoS pool size of 60% of the total available buffer memory and a headroom size of 2048 kB with priorities 1, 2, and 3 mapped to the QoS pool:

```
switch(config)# qos pool 2 lossless size 60 percent headroom 2048 kbytes  
priorities 1,2,3  
  
The setting will not be applied until configuration is saved to startup-config and  
the switch is rebooted.
```

The `no` form of the `qos pool` command will remove the configuration from the switch. For example, removing QoS pool 2:

```
switch (config)# no qos pool 2
```

PFC flow-control priorities must be removed on all interfaces prior to removing the QoS pool otherwise a warning message will be displayed and the QoS pool will not be removed from the configuration. The warning message reads as follows:

```
'flow-control priority' must be unconfigured for all interfaces before deleting pool 2.
```

Use the `no flow-control priority` command within the appropriate interface context:

```
switch(config-if)# no flow-control priority
<0-7>      The packet priority to flow control
rx         Honor received PAUSE requests for this priority
rxtx      Send and receive PAUSE requests for this priority
tx         Send PAUSE requests for this priority
```

Reapply the `no qos pool` command and the following warning message will be displayed:

```
The setting will not be applied until configuration is saved to startup-config and the switch is rebooted.
```

QoS queue profile

The QoS queue profile maps local priorities to queues. The CoS-map defines the 802.1p priority code mappings to the local priorities. The relationship to the QoS queue profile for lossless Ethernet is the CoS map and the DSCP map. The factory defaults are applied for both maps at system start up.

Default CoS map

```
switch# show qos cos-map default
code_point local_priority color  name
-----
0           1             green Best_Effort
1           0             green Background
2           2             green Excellent_Effort
3           3             green Critical_Applications
4           4             green Video
5           5             green Voice
6           6             green Internetwork_Control
7           7             green Network_Control
```

```
switch# show qos dscp-map
DSCP      code_point local_priority mpls_exp color  name
-----
000000    0           1             0       green  CS0
Codepoints 1-7 removed for brevity
001000    8           0             1       green  CS1
Codepoints 9-15 removed for brevity
010000    16          2             2       green  CS2
Codepoints 17-23 removed for brevity
```

```

011000 24      3      3      green  CS3
Codepoints 25-31 removed for brevity
100000 32      4      4      green  CS4
Codepoints 33-39 removed for brevity
101000 40      5      5      green  CS5
Codepoints 40-47 removed for brevity
110000 48      6      6      green  CS6
Codepoints 48-55 removed for brevity
111000 56      7      7      green  CS7
Codepoints 57-63 removed for brevity

```

Factory default QoS queue profile

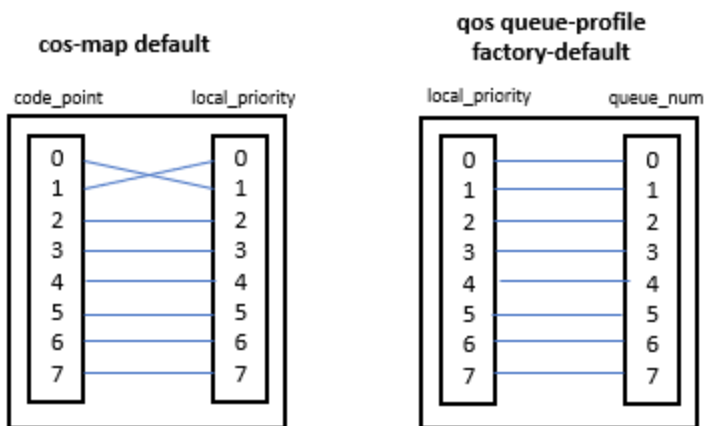
A single factory-default queue profile is initially applied to every port:

```

switch# show qos queue-profile factory-default
queue_num local_priorities name
-----
0          0                  Scavenger_and_backup_data
1          1
2          2
3          3
4          4
5          5
6          6
7          7

```

The default queue-profile has each local-priority mapped to a separate queue:



Best effort traffic will use Queue 1 using the default cos-map and qos-queue profile.

- Code point 0 (CoS value 0) is mapped to local priority 1 in the default cos-map for best effort traffic
- Code point 1 (CoS value 1) is mapped to local priority 0 in the default cos-map for less than best effort traffic
- Local priority 1 is mapped to queue 1 in the qos queue-profile factory-default
- The default CoS map and DSCP map can be leveraged noting the CoS value 0 (typically best effort traffic) will be mapped to queue 1 when using the default maps

A separate QoS queue profile is highly desirable for bandwidth allocation efficiency as it maps lossless traffic to specific and separate queues and maps lossless traffic to queues per switch hop.

QoS queue profile configuration

The preferred QoS queue profile configuration will depend on the number PFC priorities as part of the lossless traffic class and the overall QoS requirements for all traffic classes.

- A local priority can only be associated with a single queue
- Any number of local priorities can be configured to a single queue
- It is essential that local priority 7 is given a dedicated queue with a minimum bandwidth for infrastructure control traffic to avoid buffer starvation from other priority queues

QoS queue profile configuration involves the following steps:

1. Identify local priorities for mapping to the QoS queue profile
2. Map local priorities to the appropriate queue

Create a QoS queue profile called `queue-profile-1`:

```
switch(config)# qos queue-profile queue-profile-1
```

Map queues in `queue-profile-1` to local priorities:

```
switch(config)# qos queue-profile queue-profile-1
switch(config-queue)# map queue 0 local-priority 0
switch(config-queue)# map queue 1 local-priority 1,2,5,6
switch(config-queue)# map queue 2 local-priority 3
switch(config-queue)# map queue 3 local-priority 4
switch(config-queue)# map queue 4 local-priority 7
```

The `no` form of the `qos queue-profile` command will remove the QoS queue profile:

```
switch(config)# no qos queue-profile queue-profile-1
```

QoS schedule profile - enhanced transmission selection

Enhanced Transmission Selection (ETS) supports the mapping of bandwidth assignment to traffic classes using the Deficit Weighted Round Robin (DWRR) algorithm for each queue. The technique used with AOS-CX switches is to reserve bandwidth under congestion for each queue using the DWRR algorithm.

AOS-CX leverages the DWRR algorithm to provide different weights to each QoS queue which will support a single or a number of traffic classes in a profile. The profile is called the schedule-profile and by default all interfaces will have the default-schedule profile applied.

The factory default schedule profile :

```
switch# show qos schedule-profile factory-default
queue_num algorithm      weight max-bandwidth_kbps burst_KB
-----
0          dwrr          1
1          dwrr          1
2          dwrr          1
3          dwrr          1
```

4	dwrr	1
5	dwrr	1
6	dwrr	1
7	dwrr	1

DWRR Calculation

To calculate the DWRR weight relative to the interface bandwidth, the queue weight is divisible by the total number of weights for all queues. The default schedule profile has 8 queues with each queue weight of 1. To calculate the bandwidth used for each queue, the individual queue weight is divided by the total number of weights for all queues. An example using the factory default schedule profile:

- Queue 0 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 1 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 2 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 3 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 4 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 5 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 6 dwrr weight $1 - 1(8) = 12.5\%$
- Queue 7 dwrr weight $1 - 1(8) = 12.5\%$



The total percentage should add up to 100% for all weights.

For lossless Ethernet traffic flows, a custom schedule profile must provide the appropriate amount of bandwidth to the lossless queues as well as the lossy queues configured on the switch system. It must also be applied to the custom queue profile.

QoS schedule profile configuration

The following considerations should be taken into account when configuring a QoS schedule profile:

- Assign DWRR weights based on lossless flows, traffic flow volume, and a consideration to other traffic classes
- Ensure that queue 7 (assigned to critical networking traffic) has a minimum amount of bandwidth to avoid queue starvation
- The number of queues in the queue profile must align to the number of queues with allocated weights in the schedule profile
- It is recommended to use weights totaling 100 for easy reference to bandwidth percentage reservations

QoS schedule profile configuration involves the following steps:

1. Configure QoS queue profile
2. Create the schedule profile
3. Apply the QoS queue profile and schedule profile

Creating a QoS schedule profile:

```
switch (config)# qos schedule-profile schedule-profile1
```


The below example shows an ETS configuration within a 2-queue environment. The settings use a weight to set the amount of available bandwidth for each queue. The settings in the example below would ensure that 50% of bandwidth will be applied to both queue 0 and queue 1:

```
switch(config)# qos schedule-profile SMB1
switch(config-schedule)# dwrr queue 0 weight 15
switch(config-schedule)# dwrr queue 1 weight 15
```

Applying the schedule profile and queue profile:

```
switch (config)# apply qos queue-profile profile1 schedule-profile schedule-
profile1
```

The global configuration will be applied to all interfaces.

Overriding the global schedule profile on an interface

A different schedule profile can be applied directly to interfaces which will override the global schedule profile. The caveat is that the schedule profile weights must align to the same number of queues as defined in the QoS queue profile. The procedure for applying a different schedule profile to a specific interface is as follows:

1. Create a QoS queue profile that will be used by both the global schedule profile and the override schedule profile.
2. Create a QoS schedule profile and apply it to the QoS queue profile globally. This will be the schedule profile that will override the global schedule profile on the interface(s). Until this schedule profile is overridden, it will be the global schedule profile.
3. Create a schedule profile and apply it to the (same) qos queue profile. Apply the schedule-profile globally. This profile will now become the global schedule profile and will be applied to all interfaces.
4. Apply the override schedule profile to desired interface

Creating a QoS queue profile called `profile1` that will be applied to both the global schedule profile and the override interface schedule profile:

```
switch(config)# qos queue-profile profile1
switch(config-queue)# map queue 0 local-priority 0
switch(config-queue)# map queue 1 local-priority 1,2,5,6
switch(config-queue)# map queue 2 local-priority 3
switch(config-queue)# map queue 3 local-priority 4
switch(config-queue)# map queue 4 local-priority 7
```

Creating the interface override schedule profile called `profile2`:

```
switch (config)# qos schedule-profile schedule-profile2
switch (config-schedule)# dwrr queue 0 weight 5
switch (config-schedule)# dwrr queue 1 weight 50
switch (config-schedule)# dwrr queue 2 weight 20
switch (config-schedule)# dwrr queue 3 weight 15
switch (config-schedule)# dwrr queue 4 weight 10
```

Applying schedule profile `profile2` mapped with queue profile `profile1` to the global config:

```
switch(config)# apply qos queue-profile profile1 schedule-profile schedule-profile2
```

Verifying that schedule profile `profile2` has been applied to the global config:

```
switch# show qos schedule-profile  
profile_status profile_name  
-----  
complete      factory-default  
applied       schedule-profile2  
complete      strict  
incomplete     test1
```

Validating that schedule profile `profile2` is applied to all interfaces:

```
switch# show interface qos  
Interface 1/1/1 is up  
Admin state is up  
qos trust none (global)  
qos queue-profile profile1 (global)  
qos schedule-profile schedule-profile2 (global)  
  
Output removed for brevity  
  
Interface 1/1/56 is up  
Admin state is up  
qos trust none (global)  
qos queue-profile profile1 (global)  
qos schedule-profile schedule-profile2 (global)
```

Creating the schedule profile called `profile1` that will become the global schedule profile:

```
switch (config)# qos schedule-profile schedule-profile1  
switch (config-schedule)# dwrr queue 0 weight 5  
switch (config-schedule)# dwrr queue 1 weight 25  
switch (config-schedule)# dwrr queue 2 weight 35  
switch (config-schedule)# dwrr queue 3 weight 25  
switch (config-schedule)# dwrr queue 4 weight 10
```

Applying schedule profile `profile1` to queue profile `profile1`:

```
switch (config)# apply qos queue-profile profile1 schedule-profile schedule-profile1
```

Validating that the schedule profile `profile1` is applied globally and has overwritten the global setting of schedule profile `profile2`:

```
switch# show qos schedule-profile  
profile_status profile_name  
-----  
complete      factory-default  
applied       schedule-profile1  
complete      schedule-profile2
```

```
complete      strict
```

Validating the global the schedule profile `profile1` has been applied to all interfaces:

```
switch# show interface qos
Interface 1/1/1 is up
Admin state is up
qos trust none (global)
qos queue-profile profile1 (global)
qos schedule-profile schedule-profile1 (global)

output removed for brevity

Interface 1/1/56 is up
Admin state is up
qos trust none (global)
qos queue-profile profile1 (global)
qos schedule-profile schedule-profile1 (global)
```

Applying schedule profile `profile2` to a specific interface to override the global schedule profile:

```
switch (config)# interface 1/1/1
switch (config-if)# apply qos schedule-profile schedule-profile2
```

Verifying that both schedule profiles are applied:

```
switch# show qos schedule-profile
profile_status profile_name
-----
complete      factory-default
applied       schedule-profile1
applied       schedule-profile2
complete      strict
```

Verifying that schedule profile `profile2` has overridden `profile1` on interface 1/1/1 and `profile1` has been applied to all other interfaces:

```
switch# show interface qos
Interface 1/1/1 is up
Admin state is up
qos trust none (global)
qos queue-profile profile1 (global)
qos schedule-profile schedule-profile2 (override)

interfaces removed for brevity

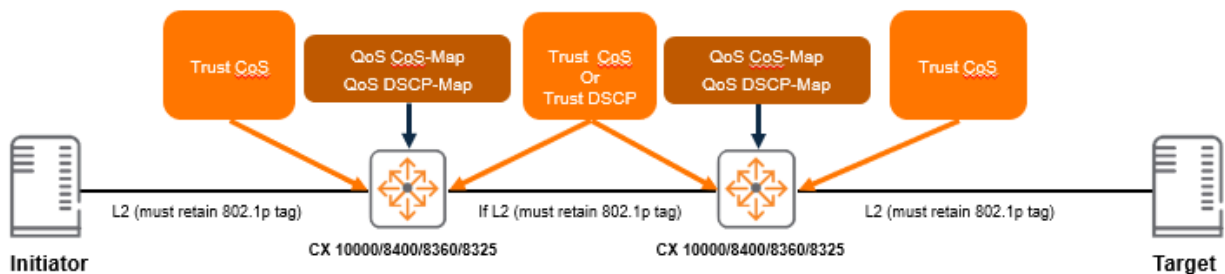
Interface 1/1/56 is up
Admin state is up
qos trust none (global)
qos queue-profile profile1 (global)
qos schedule-profile schedule-profile1 (global)
```

Schedule profile `profile2` has been applied only to interface 1/1/1. All other interfaces retain the global schedule profile setting of `profile1`.

QoS trust

QoS trust can be applied either globally or at the interface level which will override the global setting. The following considerations should be taken into account when configuring QoS trust:

- For L2 communications between switch and host the host must leverage an 802.1p tag meaning the interface between the switch and host must be a trunk link.
- For L2 switch fabric 802.1p tags must be configured between host target and initiator across the layer 2 fabric.
- For L3 switch fabric the appropriate CoS map to local preferences must be synced with the DCSP map local preferences. This is to ensure appropriate mapping for CoS and DSCP markings across the network.



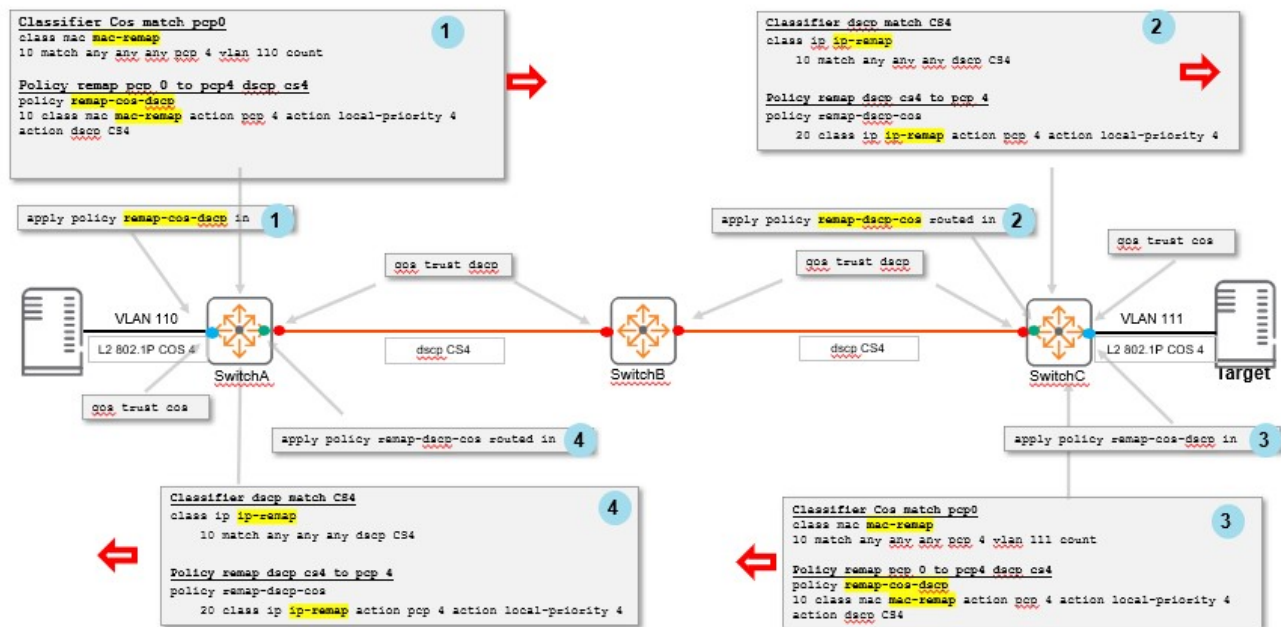
Set a global trust setting or leave QoS trust the default setting. Override the global trust setting on specific interfaces as desired.



QoS trust is set to none by default.

Layer 3 CoS-DSCP markings

Although the CoS value maybe trusted, this does not necessarily mean that the DSCP code point in the IP header is trusted from the host as it arrives on ingress on an interface from the host device. A remark policy is highly recommended to ensure that the local priority derived from the 'trusted' CoS value is remarked to the desired DSCP code point for transmission across the IP fabric. The diagram illustrates an example of layer 3 switch configuration with remark policy:



Configuration steps:

1. **SwitchA**
 - a. A classifier is applied to match traffic with a priority code point (PCP) of 4
 - b. A policy is applied to remark the classified traffic to the desired DSCP code point (CS4)
 - c. The policy is applied to the switch egress interface
2. **Switch C**
 - a. A classifier is applied to match traffic with a DSCP code point of CS4
 - b. A policy is applied to remark the classified traffic to the local-priority of 4 (CoS 4)
 - c. The policy is applied to the ingress interface (facing the initiator host)



The policies applied ensure that the DSCP value is consistent across the L3 network. At the final hop, the DSCP values is mapped to the correct CoS value for transmission to the target host. The policy is applied to the reverse flow.

3. **Switch C**
 - a. A classifier is applied to match traffic with a PCP of 4
 - b. A policy is applied to remark the classified traffic to the desired DSCP code point (CS4 in this example)
 - c. The policy is applied to the switch egress interface
4. **Switch A**
 - a. A classifier is applied to match traffic with a DSCP code point of CS4
 - b. A policy is applied to remark the classified traffic to the local-priority of 4 (CoS 4)
 - c. The policy is applied to the routed ingress interface (facing the target host)

QoS trust configuration example

```

switch# show running-config
Current configuration:
!
!
qos queue-profile profile1
    map queue 0 local-priority 0
    map queue 1 local-priority 1,5,6
    map queue 2 local-priority 2
    map queue 3 local-priority 3
    map queue 4 local-priority 4
    map queue 5 local-priority 7
qos schedule-profile schedule-profile1
    dwrr queue 0 weight 5
    dwrr queue 1 weight 15
    dwrr queue 2 weight 20
    dwrr queue 3 weight 20
    dwrr queue 4 weight 30
    dwrr queue 5 weight 10
qos schedule-profile schedule-profile2
    dwrr queue 0 weight 5
    dwrr queue 1 weight 30
    dwrr queue 2 weight 20
    dwrr queue 3 weight 25
    dwrr queue 4 weight 10
    dwrr queue 5 weight 10
apply qos queue-profile profile1 schedule-profile schedule-profile2
qos pool 1 lossless size 20.00 percent headroom 2048 kbytes priorities 2
qos pool 2 lossless size 25.00 percent headroom 2048 kbytes priorities 3
qos pool 3 lossless size 30.00 percent headroom 3072 kbytes priorities 4
interface 1/1/11
    description iSCSI
    no shutdown
    mtu 9198
    ip mtu 9198
    apply qos schedule-profile schedule-profile2
interface 1/1/12
    description NFS
    no shutdown
    mtu 9198
    flow-control priority rxtx 3
    ip mtu 9198
    apply qos schedule-profile schedule-profile2
interface 1/1/13
    description ROCEv2
    no shutdown
    mtu 9198
    flow-control priority rxtx 4
    ip mtu 9198
    apply qos schedule-profile schedule-profile1
    no shutdown
    mtu 9198
    ip mtu 9198
!
dcbx application iscsi priority 2
dcbx application tcp-udp 2049 priority 3
dcbx application udp 4791 priority 4
switch#

switch# show interface 1/1/11 qos
Interface 1/1/11 is down
Admin state is up
qos trust none (global)

```

```

qos queue-profile profile1 (global)
qos schedule-profile schedule-profile2 (override)

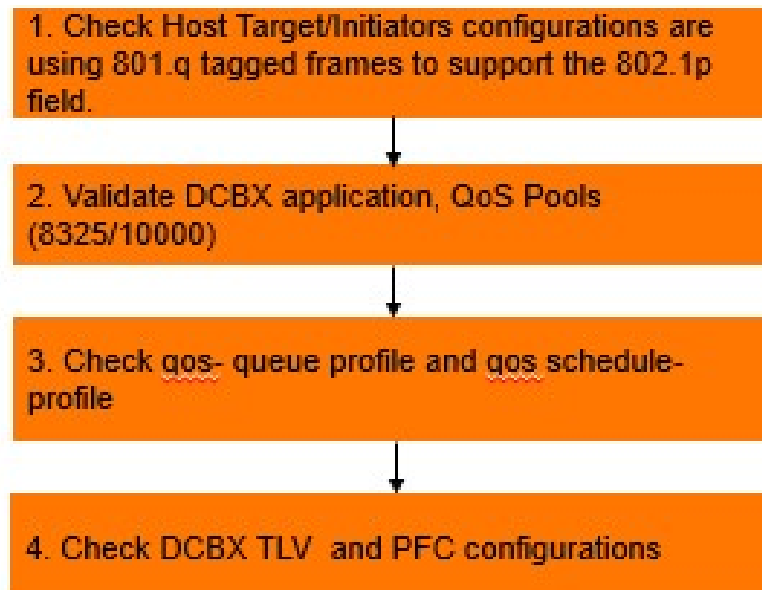
switch# show interface 1/1/12 qos
Interface 1/1/12 is down
Admin state is up
qos trust none (global)
qos queue-profile profile1 (global)
qos schedule-profile schedule-profile2 (override)

switch# show interface 1/1/13 qos
Interface 1/1/13 is down
Admin state is up
qos trust none (global)
qos queue-profile profile1 (global)
qos schedule-profile schedule-profile1 (override)

```

Troubleshooting data center bridging

It is assumed that prior to engaging in troubleshooting that host target and initiators have network connectivity, are using DCBx, and that hosts are supporting appropriate DCB protocols like PFC. Checks should be applied to every switch hop. The following flow chart represents the Aruba-recommended data center bridging troubleshooting flow supporting common DCB implementation:



1. **Verify that host target/initiator configurations are using 801.q tagged frames to support the 802.1p field.**

Check that both host and initiator are using 802.1q tagged frames between host and switch. This requires the switch interface to be configured as a trunk interface supporting the appropriate VLANs. Check that the appropriate `qos trust` setting is applied to the interface, either globally or via the interface context with the `qos trust` command. The `qos trust` setting should be set to `qos trust cos` if the host is supporting PFC as well as DCBx and is configured to use the appropriate CoS setting for a specific lossless traffic flow.

2. Validate DCBx application and QoS Pools



Only applicable to the Aruba 8325 and 10000 Switch Series

Validate that DCBx is configured on the correct interfaces, that the adjacent neighbor is using the correct DCBx version, and using the same PFC priority. If DCBx is configured and the PFC priority matches at either end for the link, the following output is displayed:

```
switch (config)# show dcbx interface 1/1/49
DCBx admin state      : enabled
DCBx operational state : active
DCBx version          : local = ieee, remote = ieee
PFC operational state  : active
```

3. Verify qos-queue profile and qos schedule-profile

The qos-queue profile is applied to the qos schedule-profile. A single qos queue-profile is applied to the system. A schedule profile must always be defined for all interfaces . Each port can have its own schedule profile.

```
switch# show qos queue-profile
profile_status profile_name
-----
complete      3qp
applied      4qp
complete      SMB
complete      factory-default
```

Verify that the correct QoS schedule profile is applied to an interface. An example of the QoS schedule profile **4qp** applied to the system:

```
switch# show qos schedule-profile
profile_status profile_name
-----
complete      3qp
applied      4qp
complete      SMB1
complete      factory-default
complete      strict
```

Verify that the correct QoS queue profile is applied. An example of the QoS queue profile **4qp** applied to the system:

4. Check DCBx TLV and PFC Configurations

Verify that the correct DCBx application TLV is configured and has the correct priority:

```
dcbx application iscsi priority 4
```

Verify that appropriate interfaces have PFC configured:

```
interface 1/1/2
no shutdown
```



```
flow-control priority rxtx 4
no routing
vlan trunk native 1
vlan trunk allowed 1,110
qos trust cos
```

DCBx commands

dcbx application

```
dcbx application {iscsi | tcp-sctp <PORT-NUM> | tcp-sctp-udp <PORT-NUM> | tcp-udp <PORT-  
NUM> | udp <PORT-NUM> | ether <ETHERTYPE>} priority <PRIORITY>  
no dcbx application
```

Description

Configures application to priority map that gets advertised in DCBx application priority messages. This tells the DCBx peer to send the application traffic with the configured priority so that the traffic is treated as lossless. Multiple applications can be configured in this manner. PFC lossless priority configured on the switch should be the same as this priority.



Priority Flow Control (PFC) commands are only supported on the 10000, 9300, 8400, 8360, and 8325 Switch Series.

The `no` form of this command removes the existing configuration.

Parameter	Description
<code>iscsi</code>	Specifies a physical port on the switch. TCP ports 860 and 3260.
<code>tcp-sctp <PORT-NUM></code>	Specifies the traffic for a specified TCP or SCTP port. Range: 1 to 65535.
<code>tcp-sctp-udp <PORT-NUM></code>	Specifies the traffic for a specified TCP or SCTP or UDP port. Range: 1 to 65535.
<code>tcp-udp <PORT-NUM></code>	Specifies the traffic for a specified TCP or UDP port. Range: 1 to 65535.
<code>udp <PORT-NUM></code>	Specifies the traffic for a specified UDP port. Range: 1 to 65535.
<code><ETHERTYPE></code>	Specifies the traffic for a specific Ethernet type. Range: 1536 to 65535.
<code><PRIORITY></code>	Specifies the application priority. Range: 0 to 7.

Usage

- In CEE DCBx version, the following traffic type configurations are sent using application TLVs:

- Ethertype
- iSCSI
- TCP-UDP
- In IEEE DCBx version, the following traffic type configurations are sent using application TLVs:
 - Ethertype
 - iSCSI
 - TCP-SCTP
 - TCP-SCTP-UDP
 - UDP

Examples

Mapping iSCSI traffic to priority 5.

```
switch(config)# dcbx application iscsi priority 5
```

Mapping TCP or SCTP traffic with port 860 to priority 3.

```
switch(config)# dcbx application tcp-sctp 860 priority 3
```



For more information on features that use this command, refer to the Fundamentals Guide for your switch model.

Command History

Release	Modification
10.08	Added a parameter option <code>tcp-udp</code> .
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	<code>config</code>	Administrators or local user group members with execution rights for this command.

lldp dcbx (global)

```
lldp dcbx [version {cee|ieee}]
no lldp dcbx [version {cee|ieee}]
```

Description

Enables advertisement of the DCBx TLVs in LLDP packets either globally or per interface. By default, DCBx is disabled in the switch.

The `no` form of this command disables DCBx advertisement.

Parameter	Description
<code>version {cee ieee}</code>	Configures the DCBx version in either CEE (Converged Enhanced Ethernet) or IEEE (IEEE 802.1Qaz). Default version: IEEE

Examples

Enabling DCBx globally with default version:

```
switch(config)# lldp dcbx
```

Disabling DCBx globally:

```
switch(config)# no lldp dcbx
```

Enabling DCBx globally with the CEE version:

```
switch(config)# lldp dcbx version cee
```

Enabling DCBx on an interface with default version:

```
switch(config-if)# lldp dcbx
```

Disabling DCBx on an interface:

```
switch(config-if)# no lldp dcbx
```

Enabling DCBx on an interface with the CEE version:

```
switch(config-if)# lldp dcbx version cee
```



For more information on features that use this command, refer to the Fundamentals Guide for your switch model.

Command History

Release	Modification
10.08	Added optional CEE and IEEE parameters.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config config-if	Administrators or local user group members with execution rights for this command.

lldp dcbx (per interface)

```
lldp dcbx [version {cee|ieee}]
no lldp dcbx [version {cee|ieee}]
```

Description

Enables DCBx on an interface. By default, an interface follows the global DCBx configuration. DCBx must be enabled globally for the interface configuration to take effect.

The `no` form of this command disables DCBx on the interface.

Parameter	Description
<code>version { cee ieee }</code>	Configures the DCBx version in either CEE (Converged Enhanced Ethernet) or IEEE (IEEE 802.1Qaz). Default version: IEEE

Usage

If the interface command specifies a different version than the global configuration, it overrides the globally configured DCBx version. If the command is executed without specifying a version, the IEEE version is configured.

Examples

Enabling DCBx on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dcbx
```

Disabling DCBx on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp dcbx
```

Enabling DCBx on interface 1/1/1 with the CEE version:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dcbx version cee
```

Enabling DCBx and configuring PFC for priority 4 on interface 1/1/1.



Priority Flow Control (PFC) commands are only supported on the 8325 and 8360.

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dcbx
switch(config-if)# flow-control priority 4
```

Command History

Release	Modification
10.08	Added a parameter 'version'
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-if	Administrators or local user group members with execution rights for this command.

lldp dcbx disable

lldp dcbx disable
no lldp dcbx disable

Description

Disables DCBx on an interface.

The `no` form of this command removes the DCBx configuration from the interface, which will then default to the global configuration.

Usage

If the interface command specifies a different version than the global configuration, it overrides the globally configured DCBx version. If the command is executed without specifying a version, the IEEE version is configured.

Examples

Disabling DCBx on interface 1/1/1:

```
switch(config)# interface 1/1/1  
switch(config-if)# lldp dcbx disable
```

Reverting interface 1/1/1 to the global DCBx configuration:

```
switch(config)# interface 1/1/1  
switch(config-if)# no lldp dcbx
```

Command History

Release	Modification
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
8360	config-if	Administrators or local user group members with execution rights for this command.

show dcbx interface

show dcbx interface <IFNAME> [peer | vsx-peer]

Description

Shows the current DCBx status and the configuration of PFC, ETS, and application priority applied on the interface and the status of the TLVs received from the peer.

Parameter	Description
interface <IFNAME>	Specifies the interface name.
peer	Shows peer DCBx information.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing DCBx on interface 1/1/1 with default DCBx version:

```
switch# show dcbx interface 1/1/1
DCBx admin state      : enabled
DCBx operational state : active
DCBx version          : local = IEEE, remote = IEEE
PFC operational state  : active

Mismatch  Advertisement          Local          Peer
-----  -
Priority Flow Control (PFC)
-> Willing:                      No             Yes
    MACsec Bypass Capability:    Yes             Yes
    Max PFC Traffic Classes:     1              1
    Priority 0:                  False            False
    Priority 1:                  False            False
    Priority 2:                  False            False
    Priority 3:                  False            False
    Priority 4:                  True             True
    Priority 5:                  False            False
    Priority 6:                  False            False
    Priority 7:                  False            False

Enhanced Transmission Selection (ETS)
-> Willing:                      No             Yes
    Credit-Based Shaper:         No             No
    Max Traffic Classes:         8              8
    Priority 0:                  Class 1          Class 1
    Priority 1:                  Class 0          Class 0
    Priority 2:                  Class 2          Class 2
    Priority 3:                  Class 3          Class 3
    Priority 4:                  Class 4          Class 4
```

Priority 5:	Class 5	Class 5
Priority 6:	Class 6	Class 6
Priority 7:	Class 7	Class 7
Class 0:	ETS 5	ETS 5
Class 1:	ETS 30	ETS 30
Class 2:	ETS 10	ETS 10
-> Class 3:	ETS 10	ETS 25
-> Class 4:	ETS 25	ETS 10
Class 5:	ETS 10	ETS 10
Class 6:	ETS 10	ETS 10
Class 7:	Strict	Strict

Application Priority Map:

Mismatch	Protocol	Protocol ID	Local Priority	Peer Priority
->	iscsi			5
	tcp-sctp	1001 (0x03E9)	1	1
->	tcp-sctp	1002 (0x03EA)	2	7
	EtherType	2000 (0x07D0)	6	6

Showing DCBx on interface 1/1/1 with CEE version:

```
switch# show dcbx interface 1/1/1
DCBx admin state      : enabled
DCBx operational state : active
DCBx version          : local = CEE, remote = CEE
PFC operational state  : active
```

Mismatch	Advertisement	Local	Peer
Control			
	Operating Version:	0	0
	Max Version:	0	0
	Sequence Number:	1	1
->	Acknowledgement Number:	1	0
Priority Flow Control (PFC)			
	Operating Version:	0	0
	Max Version:	0	0
	Enabled:	Yes	Yes
->	Willing:	No	Yes
	Error:	No	No
	Max PFC Traffic Classes:	8	8
	Priority 0:	False	False
	Priority 1:	False	False
	Priority 2:	False	False
	Priority 3:	False	False
	Priority 4:	True	True
	Priority 5:	False	False
	Priority 6:	False	False
	Priority 7:	False	False
Priority Group			
	Operating Version:	0	0
	Max Version:	0	0
->	Enabled:	Yes	No
->	Willing:	No	Yes
	Error:	No	No
	Max Traffic Classes:	8	8
	Priority 0:	PGID 1	PGID 1

```

Priority 1:                PGID 0                PGID 0
Priority 2:                PGID 2                PGID 2
Priority 3:                PGID 3                PGID 3
Priority 4:                PGID 4                PGID 4
Priority 5:                PGID 5                PGID 5
Priority 6:                PGID 6                PGID 6
Priority 7:                PGID 7                PGID 7
-> PG0 Percentage:        5                    10
-> PG1 Percentage:        10                   25
PG2 Percentage:          30                    30
PG3 Percentage:          30                    30
PG4 Percentage:          30                    30
PG5 Percentage:          30                    30
PG6 Percentage:          30                    30
PG7 Percentage:          30                    30

Application Protocol
  Operating Version:      0                    0
  Max Version:           0                    0
  Enabled:                Yes                 Yes
-> Willing:               No                  Yes
  Error:                  No                  No

Application Priority Map:
Mismatch  Protocol      Protocol ID      Local      Peer
-----  -
-> iscsi
    tcp/udp      1001 (0x03E9)    1          5
-> tcp/udp      1002 (0x03EA)    2          7
    EtherType    2000 (0x07D0)    6          6

```

Showing DCBx on interface 1/1/1 with mismatched version:

```

switch# show dcbx interface 1/1/1
DCBx admin state      : enabled
DCBx operational state : version_mismatch
DCBx version          : local = IEEE, remote = CEE
PFC operational state : active

```

Showing DCBx peer connected to an interface 1/1/1 with running CEE version:

```

switch# show dcbx interface 1/1/1 peer
DCBx version: CEE
Control
  Operating Version      : 0
  Max Version            : 0
  Sequence Number        : 1
  Acknowledgement Number : 1

Priority Flow Control (PFC)
  Operating Version      : 0
  Max Version            : 0
  Enabled                : Yes
  Willing                : No
  Error                  : No
  Max PFC Traffic Classes: 8
  Priority 0              : False
  Priority 1              : False

```



```

Priority 2           : False
Priority 3           : False
Priority 4           : True
Priority 5           : False
Priority 6           : False
Priority 7           : False

Priority Group
  Operating Version  : 0
  Max Version        : 0
  Enabled            : No
  Willing            : Yes
  Error              : No
  Max Traffic Classes : 8
  Priority 0          : PGID 1
  Priority 1          : PGID 0
  Priority 2          : PGID 2
  Priority 3          : PGID 3
  Priority 4          : PGID 4
  Priority 5          : PGID 5
  Priority 6          : PGID 6
  Priority 7          : PGID 7
  PG0 Percentage     : 25
  PG1 Percentage     : 25
  PG2 Percentage     : 10
  PG3 Percentage     : 10
  PG4 Percentage     : 5
  PG5 Percentage     : 5
  PG6 Percentage     : 10
  PG7 Percentage     : 10

```

```

Application Protocol
  Operating Version  : 0
  Max Version        : 255
  Enabled            : Yes
  Willing            : No
  Error              : No

```

```

Application Priority Map:
  Protocol      Protocol ID      Priority
  -----
  iscsi          1001 (0x03E9)    1
  tcp/udp        1002 (0x03EA)    7
  EtherType      2000 (0x07D0)    6

```



For more information on features that use this command, refer to the Fundamentals Guide for your switch model.

Command History

Release	Modification
10.08	Updated the output to show the DCBx version enhancement.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

apply qos

```
apply qos [queue-profile <QUEUE-NAME>] schedule-profile <SCHEDULE-NAME>
no apply qos [queue-profile <QUEUE-NAME>] schedule-profile <SCHEDULE-NAME>
```

Description

Applies a queue profile and schedule profile globally to all Ethernet and LAG interfaces on the switch, or applies a schedule profile to a specific interface. When applied globally, the specified schedule profile is configured only on Ethernet interfaces and LAGs that do not already have their own schedule profile.

The same profile can be applied both globally and locally to an interface. This guarantees that an interface always uses the specified profile, even if the global profile is changed.

The `no` form of this command removes the specified schedule profile from an interface and the interface uses the global schedule profile. This is the only way to remove a schedule profile override from the interface.



Interfaces may shut down briefly during reconfiguration.

Parameter	Description
<code>queue-profile <QUEUE-NAME></code>	Specifies the name of the queue profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-). This parameter is not supported in the <code>config-if</code> context.
<code>schedule-profile <SCHEDULE-NAME></code>	Specifies the name of the schedule profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

- The switch must always have a globally-applied queue and schedule profile. To stop using a given profile, apply a different profile.
- For a queue profile to be complete and ready to be applied, all eight local priorities must be mapped to a queue.
- For a schedule profile to be complete and ready to be applied, it must define all queues specified in the queue profile. All queues must use the same algorithm, except for the highest numbered queue, which can be **strict**.
- Both the queue profile and the schedule profile must specify the same number of queues.
- Schedule profiles can be modified while applied, but only in ways where a single command will not result in the profile becoming invalid. For example, queue 7 can have the algorithm changed, and weighted queues can have their weights changed.

If there are interfaces running with priority-based flow control (PFC) and a new queue profile to be applied maps a local priority used by the PFC traffic to another queue, all PFC interfaces must be shutdown before applying the new queue profile. If the new queue profile was applied before shutting down the PFC interfaces, PFC traffic will still use the same queue from the previous profile until the interfaces are shutdown then re-enabled.

If the number of queues was changed from the previous queue profile to the new one, any Ethernet or LAG interfaces with locally applied schedule profiles will program the newly applied global schedule-profile. The *show running-config interface* command will list the existing *apply qos schedule-profile* command with a comment describing the actual profile applied:

Examples

Applying the QoS profile **Q1** and the schedule profile **S1** to all interfaces that do not have an applied interface-specific schedule profile:

```
switch(config)# apply qos queue-profile Q1 schedule-profile S1
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config config-if config-lag-if	Administrators or local user group members with execution rights for this command.

apply qos threshold-profile

```
apply qos threshold-profile <THRESHOLD-NAME>  
no apply qos threshold-profile
```

Description

Applies a threshold profile globally to all Ethernet and LAG interfaces on the switch, or to a specific interface. When applied globally, the specified threshold profile is configured only on Ethernet interfaces and LAGs that do not already have their own schedule profile.

The same profile can be applied both globally and locally to an interface. This guarantees that an interface always uses the specified threshold profile, even if the global profile is changed.

The `no` form of this command removes the specified threshold profile from an interface, and causes it to use the global threshold profile. This is the only way to remove a threshold profile override from an interface. A profile can only be deleted once it is no longer applied to any interface.

Parameter	Description
<code><THRESHOLD-NAME></code>	Specifies the name of the threshold profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Example

Applying the threshold profile **mythreshold** to all interfaces that do not have an applied profile:

```
switch(config)# apply qos threshold-profile mythreshold
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config config-if config-lag-if	Administrators or local user group members with execution rights for this command.

dwrr queue

```
dwrr queue <QUEUE-NUMBER> weight <WEIGHT>
no dwrr queue <QUEUE-NUMBER>
```

Description

Assigns the deficit weighted round robin (DWRR) algorithm and its weight to a queue in a schedule profile. DWRR allocates available bandwidth among all non-empty queues in relation to the queue weights.

The **no** form of this command removes the DWRR algorithm from a queue in a schedule profile.

Parameter	Description
<code><QUEUE-NUMBER></code>	Specifies the queue number. Range: 0 to 7.
<code>weight <WEIGHT></code>	Specifies the scheduling weight. Range: 1 to 1023.

Examples

Assigning DWRR with a weight of **17** to queue **2** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# dwrr queue 2 weight 17
```

Deleting DWRR for queue **2** from the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# no dwrr queue 2
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-schedule-<NAME>	Administrators or local user group members with execution rights for this command.

map queue

```
map queue <QUEUE-NUMBER> local-priority <PRIORITY-NUMBER>
no map queue <QUEUE-NUMBER> [local-priority <PRIORITY-NUMBER>]
```

Description

Assigns a local priority to a queue in a queue profile. By default, the larger the queue number the higher its priority. A queue without a local priority value assigned to it is not used to store packets. The same queue can be assigned multiple local priorities.

The **no** form of this command removes the specified local priority from a specific queue. If no local priority number is specified, then all local priorities are removed from the queue.

Parameter	Description
<QUEUE-NUMBER>	Specifies the queue number. Range: 0 to 7.
<PRIORITY-NUMBER>	Specifies the local priority. Range: 0 to 7, where 0 is the lowest priority and 7 is the highest.

Usage

For a queue profile to be complete and ready to be applied, all eight local priorities must be mapped to a queue. Any local priority used by interface Priority-based Flow Control (PFC) must be the only local priority mapped to its queue. In order for PFC pausing to work as intended, no other local priorities

should be mapped to that same queue. This queue mapping should be configured during initial switch provisioning and only changed during maintenance periods where all ports are disabled.

The following commands illustrate a valid configuration, where every local priority value is assigned to a queue:

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 1 local-priority 2
map queue 3 local-priority 3
map queue 4 local-priority 4
map queue 5 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7
```

The following commands illustrate an invalid configuration, because local priority 2 is not assigned to a queue:

```
map queue 0 local-priority 0
map queue 1 local-priority 1
map queue 2 local-priority 3
map queue 3 local-priority 4
map queue 4 local-priority 5
map queue 5 local-priority 6
map queue 5 local-priority 7
```

Examples

Assigning priority **7** to queue **7** in profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# map queue 7 local-priority 7
```

Removing priority **7** from queue **7** in profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no map queue 7 local-priority 7
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-queue	Administrators or local user group members with execution rights for this command.

min-bandwidth

min-bandwidth queue <QUEUE-NUMBER> percent <VALUE>

no min-bandwidth queue <QUEUE-NUMBER>

Description

Assigns the Guaranteed Minimum Bandwidth (GMB) algorithm and a percentage of bandwidth to a queue. GMB allocates available bandwidth among all non-empty queues in relation to their configured minimum bandwidth. Non-empty queues are serviced first in strict order up to their minimum bandwidth. If there is any remaining bandwidth, the scheduler will strictly service any remaining non-empty queues.

The `no` form of this command only clears the algorithm for a queue if GMB has been assigned.

Parameter	Description
<QUEUE-NUMBER>	Specifies the queue number. Range: 0 to 7.
<VALUE>	Specifies bandwidth percentage used for GMB scheduling. Range: 0 to 100.

Examples

Assigning queue 0 of schedule profile S1 the GMB scheduling algorithm with minimum bandwidth of 5 percent:

```
switch(config)# qos schedule-profile S1
switch(config-schedule)# min-bandwidth queue 0 percent 5
```

Removing GMB from queue 0:

```
switch(config)# qos schedule-profile s1
switch(config-schedule)# no min-bandwidth queue 0
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-schedule-<NAME>	Administrators or local user group members with execution rights for this command.

name queue

name queue <QUEUE-NUMBER> <DESCRIPTION>

no name queue <QUEUE-NUMBER>

Description

Assigns a description to a queue in a queue profile. This is for identification purposes and has no effect on configuration.

The **no** form of this command removes the description associated with a queue.

Parameter	Description
<QUEUE-NUMBER>	Specifies the queue number. Range: 0 to 7.
<DESCRIPTION>	Specifies a queue description for identification purposes. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Examples

Assigning the description **priority-traffic** to queue **7**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# name queue 7 priority-traffic
```

Removing the description from queue **7**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no name queue 7
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-queue	Administrators or local user group members with execution rights for this command.

qos cos

```
qos cos <CODE-POINT>
no qos cos
```

Description

Configures a CoS PCP remark for an Ethernet or LAG interface. Packets that ingress on the interface are remarked at egress using the configured CoS PCP value.

The remark only occurs when QoS trust mode on the interface is set to `none`.

If QoS trust mode is not set to `none`, then the remark is ignored, and the following commands will show the CoS remark status as `ignored` (incompatible Port Access Trust configuration) or `not applied` (incompatible QoS global/port Trust configuration):

- `show running-configuration`
- `show interface <PORT-NUM>`
- `show interface <PORT-NUM> qos`

The `no` form of this command removes a CoS remark on an interface.

Parameter	Description
<code><CODE-POINT></code>	Specifies an 802.1 VLAN priority CoS value. Range: 0 to 7.

Examples

Configuring a CoS remark of **3** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# qos trust none
switch(config-if)# qos cos 3
```

Deleting a CoS remark of **3** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no qos cos
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-if	Administrators or local user group members with execution rights for this command.

qos cos-map

```
qos cos-map <CODE-POINT> local-priority <PRIORITY-NUMBER> [color <COLOR>] [name <DESCRIPTION>]
no qos cos-map <CODE-POINT>
```

Description

Defines the local priority assigned to incoming packets for a specific 802.1 VLAN priority code point (CoS) value. The CoS map values are used to mark incoming packets when QoS trust mode is set to **cos**. In **trust none** mode, CoS map entry 0 is used to set the port default local priority and color.

To see the default CoS map settings, use the following command:

```
switch# show qos cos-map default
code_point local_priority color name
-----
0          1             green Best_Effort
1          0             green Background
2          2             green Excellent_Effort
3          3             green Critical_Applications
4          4             green Video
5          5             green Voice
6          6             green Internetwork_Control
7          7             green Network_Control
```

The **no** form of this command restores the assignments for a CoS map value to the default setting.

Parameter	Description
<CODE-POINT>	Specifies an 802.1 VLAN priority CoS value. Range: 0 to 7. Default 0.
local-priority <PRIORITY-NUMBER>	Specifies a local priority value to associate with the CODE-POINT value. Range: 0 to 7. Default: 0.
color <COLOR>	Reserved for future use.
name <DESCRIPTION>	Specifies a description for the CoS setting. The name is for identification only, and has no effect on queue configuration. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

Any code point configured for use by interface Priority-based Flow Control (PFC) must be assigned a unique local priority in the CoS map. No other code point can be assigned that same local priority. This should be configured during initial switch provisioning and only changed during maintenance periods where all ports are disabled.

Examples

Mapping CoS value **1** to a local priority of **2**:

```
switch(config)# qos cos-map 1 local-priority 2
```

Mapping CoS value **1** to the default local priority value:

```
switch(config)# no qos cos-map 1
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config	Administrators or local user group members with execution rights for this command.

qos dscp

```
qos dscp <CODE-POINT>
no qos dscp
```

Description

Configures a differentiated services code point (DSCP) remark for an Ethernet or LAG interface. IPV4 and IPV6 packets that ingress on the interface are remarked at egress using the configured DSCP value.

The remark only occurs when QoS trust mode on the interface is set to `none`. If a DSCP remark is configured and then trust mode is subsequently set to `cos` or `dscp`, then the DSCP remark is ignored.

The following commands will show the remark status as *ignored* (incompatible Port Access Trust configuration) or *not applied* (incompatible QoS global or port trust configuration):

- `show running-configuration`
- `show interface <INTERFACE-NAME>`
- `show interface <INTERFACE-NAME> qos`

The `no` form of this command removes a CoS remark on an interface.

Parameter	Description
<CODE-POINT>	Specifies an IP differentiated services code point value. Range: 0 to 63.

Usage

Order of operation for arriving IPv4 or IPv6 packets:

1. Trust none is applied with initial local-priority and color metadata assigned from the CoS Map entry index 0.
2. The local-priority value and the queue profile are then used to determine the queue for the packet.
3. The remark of the packet's DSCP metadata field is performed. When the packet is transmitted, its IPv4 or IPv6 DS header is remarked with the DSCP metadata.

For arriving non-IP packets:

Trust none is applied with initial local-priority and color metadata assigned from the CoS Map entry index 0. This selects the queue for packet scheduling. The PCP of any tagged non-IP packets is unchanged.

Examples

Configuring a DSCP remark of **43** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# qos trust none
switch(config-if)# qos dscp 43
```

Deleting a DSCP remark of **43** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no dscp 43
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-if config-lag-if	Administrators or local user group members with execution rights for this command.

qos dscp-map

```
qos dscp-map <CODE-POINT> local-priority <PRIORITY-NUMBER> [color <COLOR>] [cos <PCP-VALUE>] [name <DESCRIPTION>]
no qos dscp-map <CODE-POINT>
```

Description

Defines the local priority assigned to incoming packets for a specific IP differentiated services code point (DSCP) value. The DSCP map values are used to prioritize incoming packets when QoS trust mode is set to **dscp**.

The **no** form of this command restores the assignments for a code point to the default setting.

Use **show qos dscp-map** to view the current settings. To see the default DSCP map settings, use the following command:

```
switch# show qos dscp-map default
code_point local_priority cos color name
-----
0          1              green CS0
1          1              green
2          1              green
3          1              green
4          1              green
5          1              green
...
45         5              green
46         5              green EF
47         5              green
48         6              green CS6
...
61         7              green
62         7              green
63         7              green
```

Parameter	Description
<code><CODE-POINT></code>	Specifies an IP differentiated services code point. Range: 0 to 63. Default: 0.
<code>local-priority <PRIORITY-NUMBER></code>	Specifies a local priority value to associate with the <code>CODE-POINT</code> value. Range: 0 to 7. Default: 0.
<code>color <COLOR></code>	Reserved for future use
<code>cos <PCP-VALUE></code>	Specifies an optional 802.1p VLAN Priority Code Point remark value. Range: 0 to 7. Default: No remark.
<code>name <DESCRIPTION></code>	Specifies a description for the DSCP setting. The name is used for identification only, and has no effect on queue configuration. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Examples

Setting code point **1** to a local priority of **2** and a CoS of **0**:

```
switch(config)# qos dscp-map 1 local-priority 2 cos 0
```

Setting code point **1** to the default value:

```
switch(config)# no qos dscp-map 1
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

qos queue-profile

```
qos queue-profile <NAME>
no qos queue-profile <NAME>
```

Description

Creates a new QoS queue profile and switches to the `config-queue` context for the profile. Or, if the specified QoS queue profile exists, this command switches to the `config-queue` context for the profile. A queue profile maps queues to local-priority values. Each profile has one to eight queues numbered 0 to 7. The larger the queue number, the higher its priority during transmission scheduling.

The `no` form of this command removes the specified QoS queue profile. Only profiles that are not currently applied can be removed.

Parameter	Description
<NAME>	Specifies the name of the QoS queue profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Examples

Creating the profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)#
```

Deleting the profile **myprofile**:

```
switch(config)# no qos queue-profile myprofile
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

qos schedule-profile

```
qos schedule-profile <NAME>
no qos schedule-profile <NAME>
```

Description

Creates a QoS schedule profile and switches to the `config-schedule` context for the profile. If the specified schedule profile exists, this command switches to the `config-schedule` context for the profile. The schedule profile determines the order in which queues are selected to transmit a packet, and the amount of service defined for each queue.

Parameter	Description
<NAME>	Specifies the name of the QoS schedule profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

Queues in a schedule profile are numbered consecutively starting from zero. Queue zero is the lowest priority queue. The larger the queue number, the higher priority the queue has in scheduling algorithms.

A profile named **factory-default** is defined by default and applied to all interfaces. It cannot be edited or deleted. To see its settings, use the command:

```
switch# show qos schedule-profile factory-default
queue_num algorithm weight
-----
0          dwrr        1
1          dwrr        1
2          dwrr        1
3          dwrr        1
4          dwrr        1
5          dwrr        1
```



```
6          dwrr          1
7          dwrr          1
```

A profile named **strict** is predefined and cannot be edited or deleted. The strict profile services all queues of the queue profile to which it is applied, using the strict priority algorithm.

A schedule profile must be defined on all interfaces at all times.

There are two permitted configurations for a schedule profile:

1. All queues use the same scheduling algorithm (for example, DWRR).
2. The highest queue number uses strict priority, and all remaining (lower) queues use the same algorithm (for example, DWRR). This supports priority scheduling behavior necessary for the IETF RFC 3246 Expedited Forwarding specification (<https://tools.ietf.org/html/rfc3246>).

Only limited changes can be made to an applied schedule profile:

- The weight of a dwrr queue.
- The bandwidth of a strict queue.
- The algorithm of the highest numbered queue can be swapped between dwrr and strict, and vice versa.

Applicable to REST: Any other changes will result in an unusable schedule profile, and the switch will revert to the `factory-default` profile until the profile is corrected.

The `no` form of this command removes the specified QoS schedule profile when it is not applied. Only profiles that are not currently applied to an interface can be removed.

Examples

Creating the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)#
```

Deleting the schedule profile **myschedule**:

```
switch(config)# no qos schedule-profile myschedule
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

qos shape

```
qos shape <RATE>
no qos shape
```

Description

Limits the egress bandwidth on an interface to a value that is lower than its line rate. The `no` form of this command removes shaping from an interface.

Parameter	Description
<RATE>	Specifies the maximum traffic rate in kbps. Range 64 to 100000000.

Usage

When the traffic rate destined for the port exceeds the configured egress bandwidth, the switch will buffer the excess up to the limit of the queues. Rates larger than the interface line rate will have no effect. When set on a LAG, each member Ethernet port independently shapes its egress bandwidth to the specified rate.

Examples

Configuring an egress port shaping rate of 400 Mbps on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# qos shape 400000
```

Deleting egress port shaping on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no qos shape
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-if	Administrators or local user group members with execution rights for this command.

qos threshold-profile

```
qos threshold-profile <NAME>
no qos threshold-profile <NAME>
```

Description

Creates a QoS threshold profile and switches to the `config-threshold` context for the profile. If the specified threshold profile exists, this command switches to the `config-threshold` context for the existing profile. The threshold profile determines the action to take when a threshold is exceeded for each queue.

A threshold profile is composed of up to 7 queues, numbered from 0-6. Each queue defines the action to take when buffer utilization exceeds a specific threshold.

Configure queues with the command `queue`.

The `no` form of this command removes the specified QoS threshold profile. Only profiles that are not currently applied to an interface can be removed.

Parameter	Description
<NAME>	Specifies the name of the QoS threshold profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

Queues in a threshold profile can be any valid queue number, although it is also valid to create a threshold profile with no queues specified. Queue zero is the minimum allowed queue number. The maximum allowed queue number may vary by product. For products supporting eight queues, the largest queue number is seven. If an applied threshold profile specifies configuration of a queue number that is not in use based on the configured queue profile, the threshold configuration of that unused queue is ignored.

Examples

Creating the threshold profile **mythreshold**:

```
switch(config)# qos threshold-profile mythreshold
switch(config-threshold)#
```

Deleting the threshold profile **myschedule**:

```
switch(config)# no qos threshold-profile mythreshold
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	<code>config</code>	Administrators or local user group members with execution rights for this command.

qos trust

```
qos trust {none|cos|dscp}  
no qos trust
```

Description

In the `config` context:

- This command sets the trust mode that is globally applied to all interfaces that do not have a trust mode configured.
- The `no` form of this command restores all interfaces that do not currently have a trust mode configured to the default setting.

In the `config-if` context:

- This command sets the trust mode override for a specific interface.
- The `no` form of this command clears a trust mode override. The interface then uses the global setting. This is the only way to remove a trust mode override.

Parameter	Description
<code>none</code>	Ignores all packet headers. Ingress packets are assigned the local priority and color values configured for CoS map entry 0. Default.
<code>cos</code>	For 802.1 VLAN tagged packets, use the priority code point field from the outermost VLAN header as the index into the CoS map to obtain the local priority and color values for the packet. If the packet is untagged, use the local priority and color values configured for CoS map entry 0.
<code>dscp</code>	For IP packets, use the DSCP field from the IP header as the index into the DSCP Map. For non-IP packets with 802.1 VLAN tags, use the priority code point field from the outermost VLAN header as the index into the CoS map to obtain the local priority and color values for the packet. For untagged, non-IP packets, use the local priority and color values configured for CoS map entry 0.

Example

Setting the global trust mode to **dscp**, which is applied to all interfaces that do not already have an individual trust mode configured. An override is then applied to interface **2/2/2**, and LAG 100, setting trust mode to **cos**:

```
switch(config)# qos trust dscp
switch(config)# interface 2/2/2
switch(config-if)# qos trust cos
switch(config-if)# interface lag 100
switch(config-if)# qos trust cos
```



WARNING: QoS port remark configurations are not applied when the QoS trust mode is *mode*. This warning message is seen if a port trust command other than *trust none* is attempted when there is already a remark configuration on the port. To restore the old remark configuration, configure the port trust mode to *none*.



WARNING: QoS port remark configurations are not applied when the global QoS trust mode is *mode*. This warning message is seen if a port *no qos trust* command is attempted when there is already a remark configuration on the port and the global trust mode is not *none*. To re-apply the remark configuration, set the port trust mode to *none*.



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config config-if config-lag-if	Administrators or local user group members with execution rights for this command.

queue action

8360 Switch Series

```
queue <0-7> action ecn all threshold <LIMIT> percent
no queue <0-7>
```

Description

Defines the threshold settings and action for a specified queue in a threshold-profile. When queue utilization exceeds the threshold value, ECT (ECN-Capable Transport) packets will be CE (Congestion Encountered) marked when transmitted.

The **no** form of this command removes the settings for a queue.

Parameter	Description
<0-7>	Specifies the queue number. Range: 0 to 7.
<LIMIT>	Specifies the threshold value in percentage. Range: 0 to 100 percentage.

Examples

Assigning a threshold to queue **7** in profile **mythreshold**:

```
switch(config)# qos threshold-profile mythreshold
switch(config-threshold)# queue 7 action ecn all threshold 70 percent
```

Removing a threshold from queue **7** in profile **mythreshold**:

```
switch(config)# qos threshold-profile mythreshold
switch(config-threshold)# no queue 7
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-threshold	Administrators or local user group members with execution rights for this command.

rate-limit

```
rate-limit {broadcast | multicast | unknown-unicast | icmp {ip-all | ipv4 | ipv6}} <RATE>
{kbps | pps}
no rate-limit {broadcast | multicast | icmp}
```

Description

Sets the amount of traffic of a specific type that can ingress on an Ethernet interface, or on each port of a LAG interface. Rate limits are enforced separately on each individual member of a LAG, not on the LAG as a whole.

The **no** form of this command removes the traffic limit for the specified traffic type.

Parameter	Description
<code>{broadcast multicast icmp {ip-all ipv4 ipv6}}</code>	Specifies the type of ingress traffic to which the rate limit applies: broadcast, multicast, or ICMP. The multicast rate limit affects multicast and broadcast traffic. The broadcast rate limit only affects broadcast traffic. When both types are applied to the same interface, broadcast packets are limited to the lower of the two rate values. Layer 2 BPDU packets, like spanning tree, are also included in the multicast rate limit. The ICMP rate limit can be configured to apply to IPv4, IPv6, or all IP traffic. Only one ICMP rate-limit can be configured at a time. Applying a new ICMP rate-limit replaces any previous ICMP rate-limit.
<code><RATE> {kbps pps}</code>	Specifies the rate limit. Range: 64 to 100000000 kbps (in steps of 64 kbps), or 64 to 209090910 kbps (in steps of 64 pps). The actual rate limit varies with steps approximately equal to the minimum value. Verify the actual rate limit using the command <code>show interface <INTERFACE-NAME></code> .

Examples

Limiting broadcast traffic to **2000pps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit broadcast 500 kbps
```

Limiting all ICMP IPv4 traffic to **10000kbps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit icmp ipv4 10000 kbps
```

Viewing the results of the previous configuration settings:

```
switch# show interface 1/1/3 qos
Interface 1/1/3 is up
Admin state is up
Description:
Hardware: Ethernet, MAC Address: 08:97:34:b1:20:00
MTU 1500
Type 1000BT
qos trust none
rate-limit broadcast 2000 pps
(2000 actual)
rate-limit icmp ip-all 10000 kbps
```

```

(10000 actual)
Speed 1000 Mb/s
L3 Counters: Rx Disabled, Tx Disabled
Auto-Negotiation is on
Flow-control: off
Rx
    0 input packets          0 bytes
    0 input error            0 dropped
    0 CRC/FCS
L3:
    0 packets, 0 bytes
Tx
    127 output packets       16510 bytes
    0 input error             0 dropped
    0 collision
L3:
    0 packets, 0 bytes

```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-if	Administrators or local user group members with execution rights for this command.

rate-limit

```

rate-limit {broadcast | multicast | unknown-unicast} <RATE> pps
no rate-limit {broadcast | multicast | unknown-unicast}

```

Description

Sets the amount of traffic of a specific type that can ingress on an Ethernet port, or on each port of a LAG interface. Rate limits are enforced separately on each individual member of a LAG, not on the LAG as a whole.

The `no` form of this command removes the traffic limit for the specified traffic type.

Parameter	Description
{broadcast multicast unknown-unicast}	Specifies the type of ingress traffic to which the rate limit applies: <code>broadcast</code> , <code>multicast</code> , or <code>unknown-unicast</code> . The multicast rate limit affects multicast and broadcast traffic. The broadcast rate limit only

Parameter	Description
	affects broadcast traffic. When both types are applied to the same interface, broadcast packets are limited to the lower of the two rate values. Layer 2 BPDU packets, like spanning tree, are also included in the multicast rate limit.
<i>rate</i>	Specifies the rate limit in packets per second (pps). Range: 64 to 209090910 pps. The actual rate limit varies with steps approximately equal to the minimum value. Verify the actual rate limit using the command <code>show interface <INTERFACE-NAME></code> .

Examples

Limiting multicast traffic to 4000 pps on interface 1/3/3:

```
switch(config)# interface 1/3/3
switch(config-if)# rate-limit multicast 4000 pps
```

Viewing the results of the previous configuration setting:

```
switch# show interface 1/3/3 qos
Interface 1/3/3 is down (Administratively down)
Admin state is down
Hardware: Ethernet, MAC Address: 1c:98:ec:e3:6a:00
MTU 1500
Full-duplex
rate-limit multicast 4000 pps (4000 actual)

Speed 0 Mb/s
Auto-Negotiation is turned on
Input flow-control is off, output flow-control is off
RX
    0 input packets          0 bytes
    0 input error            0 dropped
    0 CRC/FCS
L3:
    ucast: 0 packets, 0 bytes
    mcast: 0 packets, 0 bytes
TX
    0 output packets         0 bytes
    0 input error            0 dropped
    0 collision
L3:
    ucast: 0 packets, 0 bytes
    mcast: 0 packets, 0 bytes
```

Limiting broadcast traffic to 50 pps on LAG 100:

```
switch# config
switch(config)# interface 1/3/3
switch(config-if)# interface lag 100
switch(config-if)# rate-limit broadcast 50 pps
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	config-if	Administrators or local user group members with execution rights for this command.

show interface queues

show interface <INTERFACE-NAME> queues

Description

Displays interface-level queue statistics.

Parameter	Description
<INTERFACE-NAME>	Specifies the name of an Ethernet port or LAG on the switch. Format: member/slot/port or lag number.

Usage

Statistics include:

- **Tx Bytes:** Total bytes transmitted. The byte count may include packet headers and internal metadata that are removed before the packet is transmitted. Packet headers added when the packet is transmitted may not be included. The byte count includes any packets subsequently dropped by an egress ACL .
- **Tx Packets:** Total packets transmitted. The count includes packets subsequently dropped by an egress ACL .
- **Tx Drops:** Total packets dropped by an egress queue due to insufficient capacity.

Examples

Showing queue statistics for interface **1/1/5**:

```
switch# show interface 1/1/5 queues
Interface 1/1/5 is down
Admin state is up
      Tx Bytes      Tx Packets      Tx Drops
Q0              0             0             3
Q1          15356             73             0
```

Q2	0	0	0
Q3	0	0	0
Q4	0	0	0
Q5	0	0	0
Q6	0	0	0
Q7	0	0	0

Showing queue statistics for interface **lag 1**:

```
switch# show interface lag 1 queues
Aggregate-name lag1
Aggregated-interfaces :
1/1/6 1/1/7
Speed 20000 Mb/s
      Tx Bytes      Tx Packets      Tx Drops
Q0              0              0              0
Q1              0              0              0
Q2              0              0              0
Q3              0              0              0
Q4              0              0              0
Q5              0              0              0
Q6              0              0              0
Q7            3450             25              0
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface qos

show interface <INTERFACE-NAME> qos

Description

Shows various QoS settings for a specific interface.

Parameter	Description
<INTERFACE-NAME>	Specifies the name of an interface on the switch. Format: member/slot/port or lag number.

Examples

Showing QoS settings for interface 1/1/5:

```
switch# show interface 1/1/5 qos
Interface 1/1/5 is down
Admin state is up
qos trust none (global)
qos queue-profile factory-default (global)
qos schedule-profile EQSExample (override)
qos threshold-profile DefaultThresh (global)
qos dscp 47
rate-limit broadcast 500 pps (500 actual)
rate-limit multicast 500 pps (500 actual)
Forwarded Pkts          Dropped Pkts          Forwarded Bytes          Dropped Bytes
Broadcast:              944468              1044              1044658890
                        85662408
Multicast:              20002              3041560              29756804
                        9465800238
qos shape 200000 (199999 actual)
Max-Bandwidth Kbps
Q1              10082461
Q4              20164923
Q7              30247384
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show qos cos-map

```
show qos cos-map [default] [vsx-peer]
```

Description

Shows the global QoS CoS code point settings, or the factory default settings.

Parameter	Description
default	Shows the factory default CoS code point settings.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing the current CoS map:

```
switch# show qos cos-map
code_point  local_priority  color  name
-----
0           2              green  Best_Effort
1           0              green  Background
2           1              green  Spare
3           3              green  Excellent_Effort
4           4              green  Controlled_Load
5           5              green  Video
6           6              green  Voice
7           7              green  Network_Control
```

Showing the default CoS map:

```
switch# show qos cos-map default
code_point  local_priority  color  name
-----
0           1              green  Best_Effort
1           0              green  Background
2           2              green  Excellent_Effort
3           3              green  Critical_Applications
4           4              green  Video
5           5              green  Voice
6           6              green  Internetwork_Control
7           7              green  Network_Control
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show qos dscp-map

show qos dscp-map [default] [vsx-peer]

Description

Displays the current or default global QoS dscp-map.

Parameter	Description
default	Shows the factory default DSCP code point settings.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing the current QoS DSCP map:

```
switch# show qos dscp-map
code_point  local_priority  cos  color  name
-----
0           1              3    green  CS0
1           1              3    green
2           1              3    green
3           1              3    green
4           1              3    green
5           1              3    green
6           1              3    green
7           1              3    green
8           0              3    green  CS1
...
45          5              7    green
46          7              7    green  EF
47          5              7    green
48          6              7    green  CS6
...
61          7              7    green
62          7              7    green
63          7              7    green
```

Showing the default QoS DSCP map:

```
switch# show qos dscp-map default
code_point  local_priority  cos  color  name
-----
0           1              3    green  CS0
1           1              3    green
2           1              3    green
3           1              3    green
4           1              3    green
5           1              3    green
...
45          5              7    green
46          5              7    green  EF
47          5              7    green
48          6              7    green  CS6
```

```
...
61      7      green
62      7      green
63      7      green
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show qos queue-profile

```
show qos queue-profile [<NAME> | factory-default] [vsx-peer]
```

Description

Shows the status of all queue profiles, or a specific queue profile.

Parameter	Description
<NAME>	Specifies the name of a queue profile. Range 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).
[factory-default]	Specifies the factory default queue profile.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

The status of a queue profile can be:

- Applied - The profile is actively being used by the switch.
- Complete - The profile meets the criteria to be applied.
- Incomplete - The profile does not meet the criteria to be applied.

For a queue profile to be complete and ready to be applied:

- All eight local priorities must be mapped to some queue.
- There can be 1 to 8 queues.
- The queues must be consecutively numbered starting at zero.

Examples

Showing the settings of the factory default queue profile:

```
switch# show qos queue-profile factory-default
queue_num local_priorities name
-----
0          0
1          1
2          2
3          3
4          4
5          5
6          6
7          7
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show qos schedule-profile

show qos schedule-profile [<NAME> | factory-default | strict] [vsx-peer]

Description

Shows the status of all schedule profiles, or a specific schedule profile.

Parameter	Description
<NAME>	Specifies the name of a queue or schedule profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Parameter	Description
[factory-default]	Specifies the factory default queue profile.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

The status of a schedule profile can be:

- Applied - The profile is actively being used by one or more ports.
- Complete - The profile meets the criteria to be applied.
- Incomplete - The profile does not meet the criteria to be applied.

For a schedule profile to be complete and ready to be applied it must have:

- An algorithm for each queue defined by the applied queue profile.
- All queues must use the same algorithm except for the highest numbered queue, which may be strict.

Example

Showing the status of all schedule profiles:

```
switch# show qos schedule-profile
profile_status profile_name
-----
applied        myschedule
complete       factory-default
complete       Test
```

Showing the configuration of factory default schedule profile:

```
switch# show qos schedule-profile factory-default
queue_num algorithm weight
-----
0          dwrr         1
1          dwrr         1
2          dwrr         1
3          dwrr         1
4          dwrr         1
5          dwrr         1
6          dwrr         1
7          dwrr         1
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show qos threshold-profile

show qos threshold-profile [<NAME> [vsx-peer]

Description

Shows the status of all threshold profiles, or a specific threshold profile.

Parameter	Description
<NAME>	Specifies the name of a threshold profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

Status fields are:

- applied: The threshold profile is applied to all configured ports.
- partially applied: The threshold profile is applied to some configured ports but failed on other configured ports.
- not applied: The threshold profile could not be applied to any configured ports.
- blank field: The threshold profile is not applied in the configuration (globally or as a port override).

Examples

Showing the status of all threshold profiles:

```
switch# show qos threshold-profile
profile_status      profile_name
-----
CustomThresh
applied             mythreshold
```

Showing the status of threshold profile **mythreshold**:

On the 8360 switches:

```

switch# show qos threshold-profile mythreshold
queue_num action Threshold
-----
5          ecn          40
7          ecn          50

Ports      Status
-----
1/1/1      applied
1/1/8      applied
1/2/2      applied

```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.08	Updated the example.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8360	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show qos trust

```
show qos trust [default] [vsx-peer]
```

Description

Shows the global QoS trust settings, or the factory default settings.

Parameter	Description
default	Shows the factory default QoS trust settings.
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

Showing the current QoS trust settings:

```
switch# show qos trust
qos trust cos
```

Showing the default QoS trust settings:

```
switch# show qos trust default
qos trust none
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

strict queue

```
strict queue <QUEUE-NUMBER> [[max-bandwidth <BANDWIDTH>]]
no strict queue <QUEUE-NUMBER>
```

Description

Assigns the strict priority algorithm to a queue. Strict priority services all packets waiting in a queue, before servicing the packets in lower priority queues.

Egress queue shaping can be configured using the `max-bandwidth` option to limit the amount of traffic transmitted per output queue. The buffer associated with each egress queue stores the excess traffic to smooth the output rate. Sustained rates of traffic above the maximum bandwidth will eventually fill the output queue causing tail drops. Use the command `show interface` to determine if any tail drop errors have occurred.

The `no` form of this command removes the queue configuration from the schedule profile. To remove only egress queue shaping, re-enter the `strict queue` command without the `max-bandwidth` parameter.

Parameter	Description
<QUEUE-NUMBER>	Specifies the number of the queue. Range: 0 to 7.
max-bandwidth <BANDWIDTH>	Specifies the maximum bandwidth allowed on the queue in Kbps. Range: 64 to 100000000.

Usage

Either all the queues of the schedule profile can be *strict* or just the highest numbered queue. When applied to a LAG, each member Ethernet port independently schedules its egress transmissions using the strict settings. Only limited changes can be made to a *strict* queue that is part of an applied schedule profile:

- The max-bandwidth settings.
- The highest numbered queue can be swapped between *strict* and *dwrr* or *min-bandwidth*

Any other changes or removing a queue (*no strict queue*) will result in an unusable schedule profile. If that schedule profile is applied in the interface context, the switch will revert to the schedule profile applied in the global context until the profile is corrected. If that schedule profile is applied in the global context, the switch will revert to using the factory-default profile until the profile is corrected.

Examples

Assigning strict priority to queue **7** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7
```

Deleting strict priority from queue **7** in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# no strict queue 7
```

Assigning strict priority to queue **7** in the schedule profile **myschedule** with a maximum bandwidth of 10000 Kbps:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7 max-bandwidth 10000
```



For more information on features that use this command, refer to the Quality of Service Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-schedule- <i><NAME></i>	Administrators or local user group members with execution rights for this command.

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.arubanetworks.com/support-services/
AOS-CX Switch Software Documentation Portal	https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

channel on
YouTube.

HPE Aruba
Networking
Hardware
Documentation
and Translations
Portal

<https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm>

HPE Aruba
Networking
software

<https://networkingsupport.hpe.com/downloads>

Software
licensing and
Feature Packs

<https://licensemanagement.hpe.com/>

End-of-Life
information

<https://www.arubanetworks.com/support-services/end-of-life/>

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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